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CSA5801- DevSecOps Engineering and Application Security

SLOT-A

LAB PROGRAM

CODES WITH THEIR OUTPUT ;

```
# =====
```

```
# CSA58 DevSecOps Engineering and Application Security
```

```
# 40 R PROGRAMS
```

```
# Rule:
```

```
# Static / workflow / design / modelling -> DiagrammeR
```

```
# Dynamic / simulation / analysis / results -> dygraphs
```

```
# =====
```

```
# Install once:
```

```
# install.packages("DiagrammeR")
```

```
# install.packages("dygraphs")
```

```
# install.packages("xts")
```

```
library(DiagrammeR)
```

```
library(dygraphs)
```

```
library(xts)
```

```
# =====
```

```
# EXPERIMENT 1
```

```
# Installation and Configuration of Git, GitHub, RStudio
```

```
# and Required DevSecOps Tools
```

```
# =====
```

```
library(DiagrammeR)
```

```
grViz("
```

```
digraph DevSecOpsSetup {
```

```
graph [
```

```
  rankdir = TB,
```

```
  bgcolor = white,
```

```
  labelloc = t,
```

```
  label = 'DevSecOps Tool Installation and Configuration',
```

```
  fontsize = 20
```

```
]
```

```
node [
```

```
  shape = box,
```

```
  style = 'rounded,filled',
```

```
  fontname = Helvetica,
```

```
  fontsize = 14
```

```
]
```

```
Git [
```

```
  label = '1. Install Git',
```

```
  fillcolor = lightblue
```

```
]
```

```
GitHub [
```

```
  label = '2. Create / Configure GitHub',
```

```
  fillcolor = lightgreen
```

```
]
```

```
RStudio [  
  label = '3. Install RStudio',  
  fillcolor = lightyellow  
]
```

```
R [  
  label = '4. Configure R Environment',  
  fillcolor = lightpink  
]
```

```
Tools [  
  label = '5. Configure DevSecOps Tools',  
  fillcolor = lightcyan  
]
```

```
Verify [  
  label = '6. Verify Installation',  
  shape = diamond,  
  fillcolor = lightgray  
]
```

```
Ready [  
  label = 'DevSecOps Environment Ready',  
  shape = oval,  
  fillcolor = lightgreen  
]
```

Git -> GitHub

GitHub -> RStudio

RStudio -> R

R -> Tools

Tools -> Verify

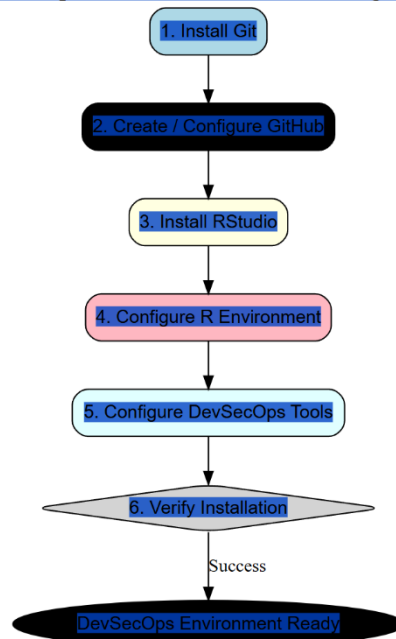
Verify -> Ready [label = 'Success']

}

")

Viewer Zoom

DevSecOps Tool Installation and Configuration



```
# =====  
# EXPERIMENT 2  
# Visualization of the DevSecOps Lifecycle using Workflow Modelling  
# =====
```

```
# Run this once if DiagrammeR is not installed  
install.packages("DiagrammeR")
```

```
# Load package  
library(DiagrammeR)
```

```
# DevSecOps lifecycle workflow  
grViz("  
digraph DevSecOps {  
  
graph [  
  rankdir=LR,  
  bgcolor=white,  
  label='DevSecOps Lifecycle Workflow',  
  labelloc=t,  
  fontsize=20  
]  
  
node [  
  shape=box,  
  style='rounded, filled',  
  fontname=Helvetica,  
  fontsize=13  
]
```

```
Plan [  
    label='PLAN\\nRequirements & Security Planning',  
    fillcolor=lightblue  
]
```

```
Code [  
    label='CODE\\nSecure Coding',  
    fillcolor=lightgreen  
]
```

```
Build [  
    label='BUILD\\nCompile & Package',  
    fillcolor=lightyellow  
]
```

```
Test [  
    label='TEST\\nFunctional & Security Testing',  
    fillcolor=lightpink  
]
```

```
Security [  
    label='SECURITY\\nSAST / DAST / SCA',  
    fillcolor=lightcyan  
]
```

```
Deploy [  
    label='DEPLOY\\nSecure Deployment',  
    fillcolor=lavender  
]
```

```
]
```

```
Monitor [
```

```
    label='MONITOR\\nSecurity Monitoring',
```

```
    fillcolor=wheat
```

```
]
```

```
Feedback [
```

```
    label='FEEDBACK\\nRisk & Improvement',
```

```
    shape=diamond,
```

```
    fillcolor=lightgray
```

```
]
```

```
Plan -> Code
```

```
Code -> Build
```

```
Build -> Test
```

```
Test -> Security
```

```
Security -> Deploy
```

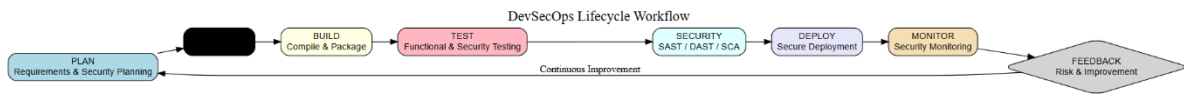
```
Deploy -> Monitor
```

```
Monitor -> Feedback
```

```
Feedback -> Plan [label='Continuous Improvement']
```

```
}
```

```
")
```




```
# =====  
# EXPERIMENT 3  
# Comparative Analysis of DevOps and DevSecOps using Synthetic Data  
# =====
```

```
# Run once if required
```

```
install.packages("dygraphs")
```

```
install.packages("xts")
```

```
# Load packages
```

```
library(dygraphs)
```

```
library(xts)
```

```
# Synthetic performance data
```

```
days <- as.Date("2026-07-20") + 0:9
```

```
data <- xts(  
  cbind(  
    DevOps = c(62, 64, 66, 65, 68, 70, 71, 72, 74, 75),  
    DevSecOps = c(60, 64, 69, 72, 75, 78, 81, 84, 87, 90)  
  ),  
  order.by = days  
)
```

```
# Interactive comparison graph
```

```
dygraph(  
  data,
```

```
  data,
```

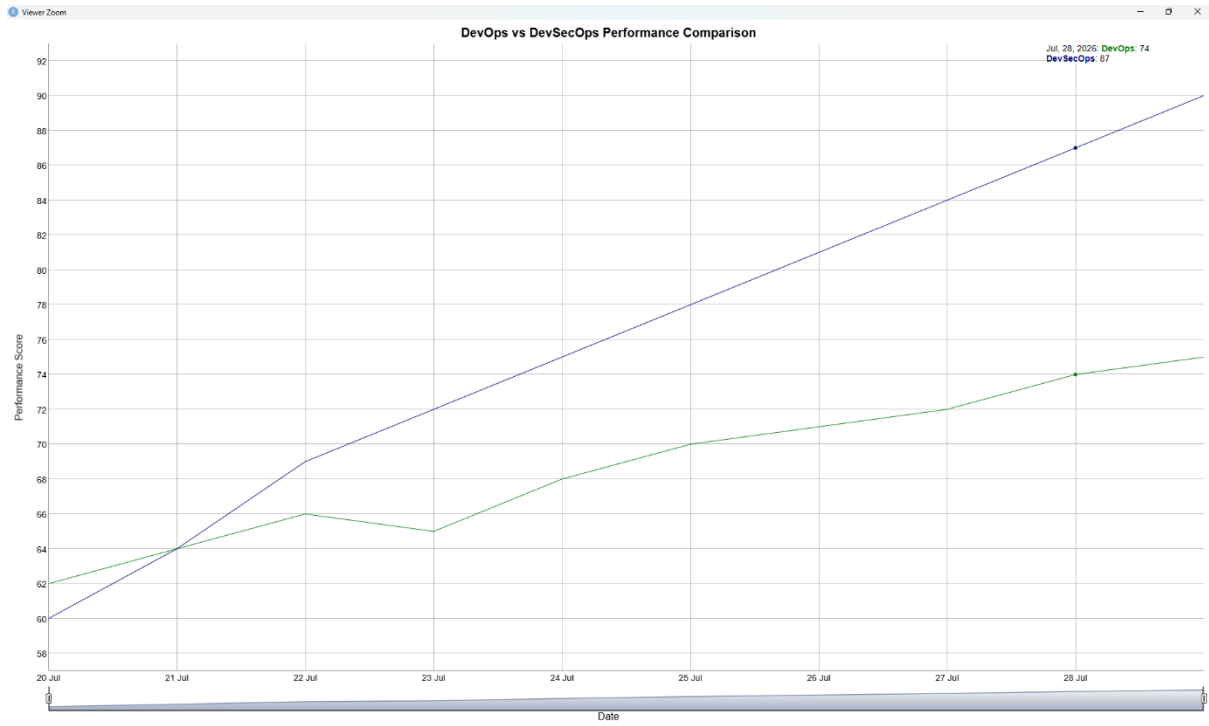
```
  main = "DevOps vs DevSecOps Performance Comparison",
```

```
  xlab = "Date",
```

```
ylab = "Performance Score"
```

```
) %>%
```

```
dyRangeSelector()
```

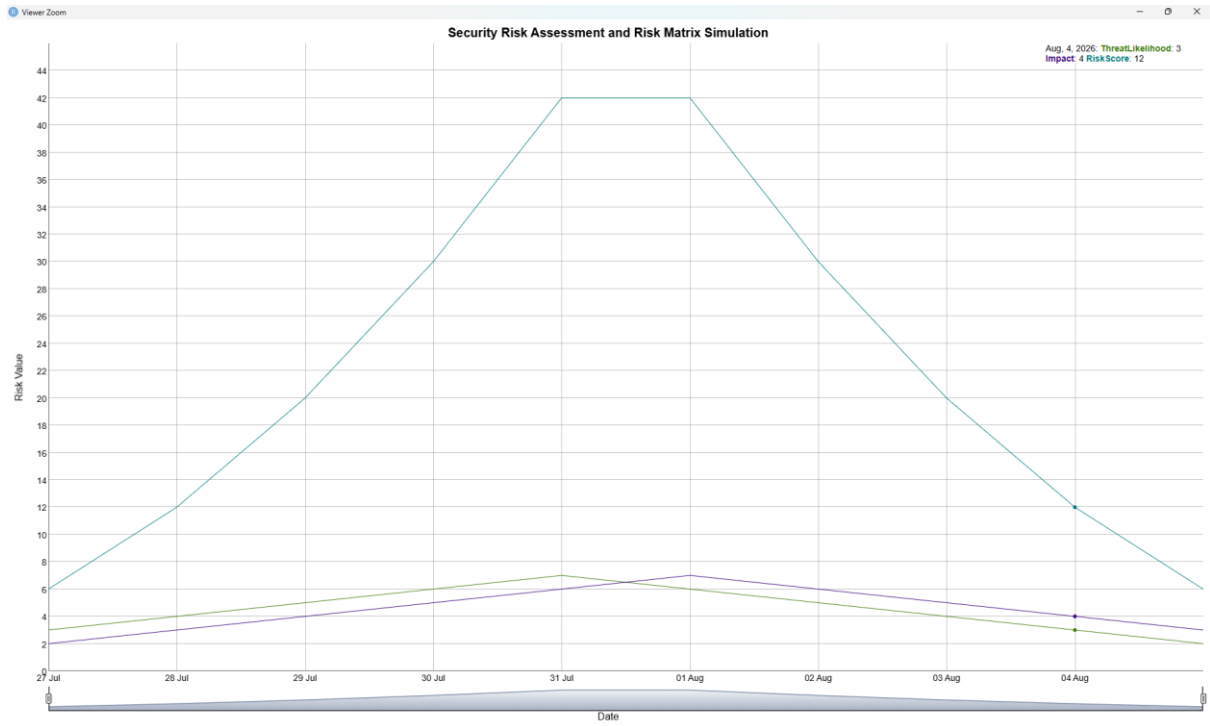


```
# =====  
# EXPERIMENT 4  
# Security Risk Assessment and Risk Matrix Simulation  
# =====  
  
# Run once if required  
install.packages("dygraphs")  
install.packages("xts")  
  
library(dygraphs)  
library(xts)  
  
# Synthetic security risk data  
days <- as.Date("2026-07-27") + 0:9  
  
risk_data <- xts(  
  cbind(  
    ThreatLikelihood = c(3, 4, 5, 6, 7, 6, 5, 4, 3, 2),  
    Impact = c(2, 3, 4, 5, 6, 7, 6, 5, 4, 3),  
    RiskScore = c(6, 12, 20, 30, 42, 42, 30, 20, 12, 6)  
  ),  
  order.by = days  
)  
  
# Interactive security risk simulation  
dygraph(  
  risk_data,  
  main = "Security Risk Assessment and Risk Matrix Simulation",  
  xlab = "Date",
```

```
ylab = "Risk Value"
```

```
) %>%
```

```
dyRangeSelector()
```



```
# =====  
# EXPERIMENT 5  
# Threat Propagation Simulation and Attack Surface Analysis  
# =====
```

```
# Run once if required
```

```
install.packages("dygraphs")
```

```
install.packages("xts")
```

```
library(dygraphs)
```

```
library(xts)
```

```
# Synthetic threat propagation data
```

```
days <- as.Date("2026-08-03") + 0:9
```

```
threat_data <- xts(
```

```
  cbind(
```

```
    ThreatLevel = c(2, 4, 7, 11, 16, 14, 11, 8, 5, 3),
```

```
    AttackSurface = c(10, 14, 19, 26, 34, 31, 27, 22, 17, 13),
```

```
    ExposedAssets = c(3, 5, 8, 12, 16, 14, 11, 8, 5, 4)
```

```
  ),
```

```
  order.by = days
```

```
)
```

```
# Interactive threat propagation analysis
```

```
dygraph(
```

```
  threat_data,
```

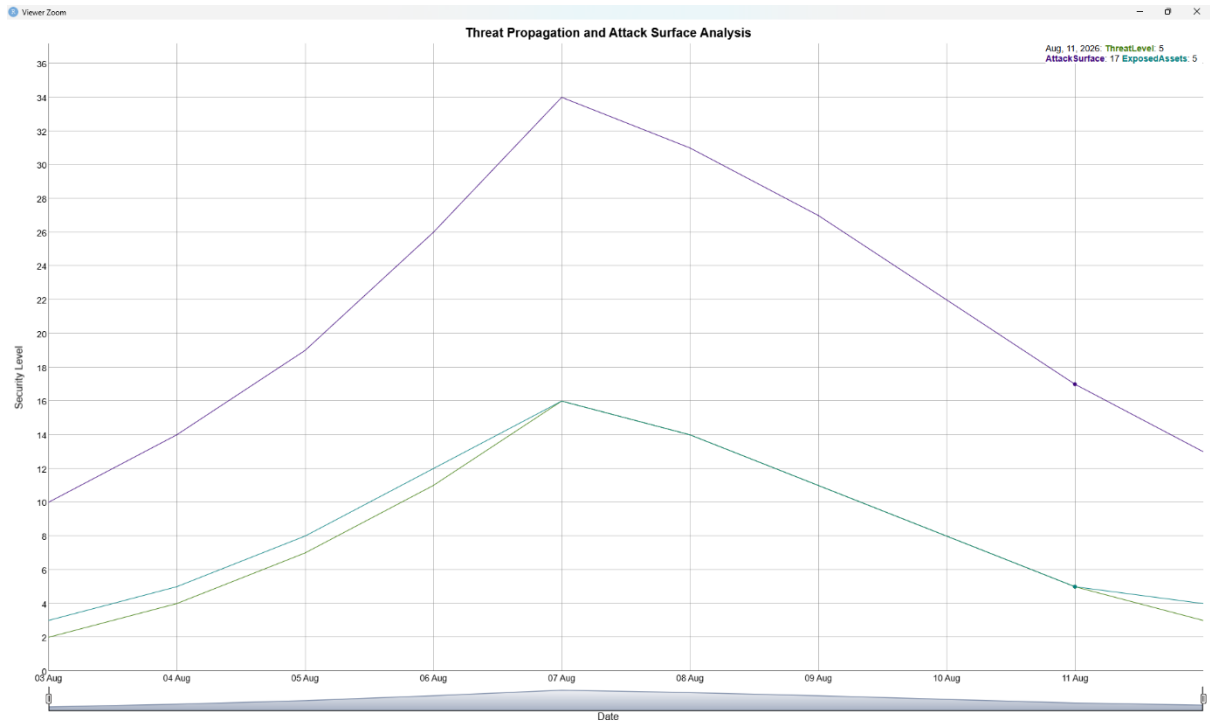
```
  main = "Threat Propagation and Attack Surface Analysis",
```

```
  xlab = "Date",
```

ylab = "Security Level"

) %>%

dyRangeSelector()



```
# =====
```

```
# EXPERIMENT 6
```

```
# Design and Simulation of Shift-Left Security Workflow
```

```
# =====
```

```
# Run once if required
```

```
install.packages("DiagrammeR")
```

```
library(DiagrammeR)
```

```
grViz("
```

```
digraph ShiftLeftSecurity {
```

```
graph [
```

```
  rankdir=LR,
```

```
  bgcolor=white,
```

```
  label='Shift-Left Security Workflow',
```

```
  labelloc=t,
```

```
  fontsize=20
```

```
]
```

```
node [
```

```
  shape=box,
```

```
  style='rounded, filled',
```

```
  fontname=Helvetica,
```

```
  fontsize=13
```

```
]
```

```
Plan [
```

```
    label='Security Planning',  
    fillcolor=lightblue  
]
```

```
Code [  
    label='Developer Writes Code',  
    fillcolor=lightgreen  
]
```

```
SAST [  
    label='Early Security Check\\n(SAST)',  
    fillcolor=lightyellow  
]
```

```
Fix [  
    label='Fix Security Issues',  
    fillcolor=lightpink  
]
```

```
Test [  
    label='Security Testing',  
    fillcolor=lightcyan  
]
```

```
Build [  
    label='Secure Build',  
    fillcolor=lavender  
]
```



```
Deploy [  
  label='Deploy',  
  fillcolor=wheat  
]
```

Plan -> Code

Code -> SAST

SAST -> Fix [label='Issues Found']

Fix -> SAST [label='Recheck']

SAST -> Test [label='No Critical Issues']

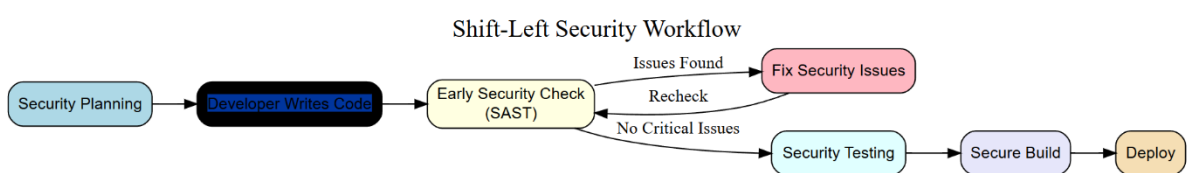
Test -> Build

Build -> Deploy

}

")

Viewer Zoom



```
# =====  
# EXPERIMENT 7  
# Static Application Security Testing (SAST)  
# and Security Verification  
# =====
```

```
# Run once if required  
install.packages("dygraphs")  
install.packages("xts")
```

```
library(dygraphs)  
library(xts)
```

```
# Synthetic SAST scan results  
days <- as.Date("2026-08-10") + 0:9
```

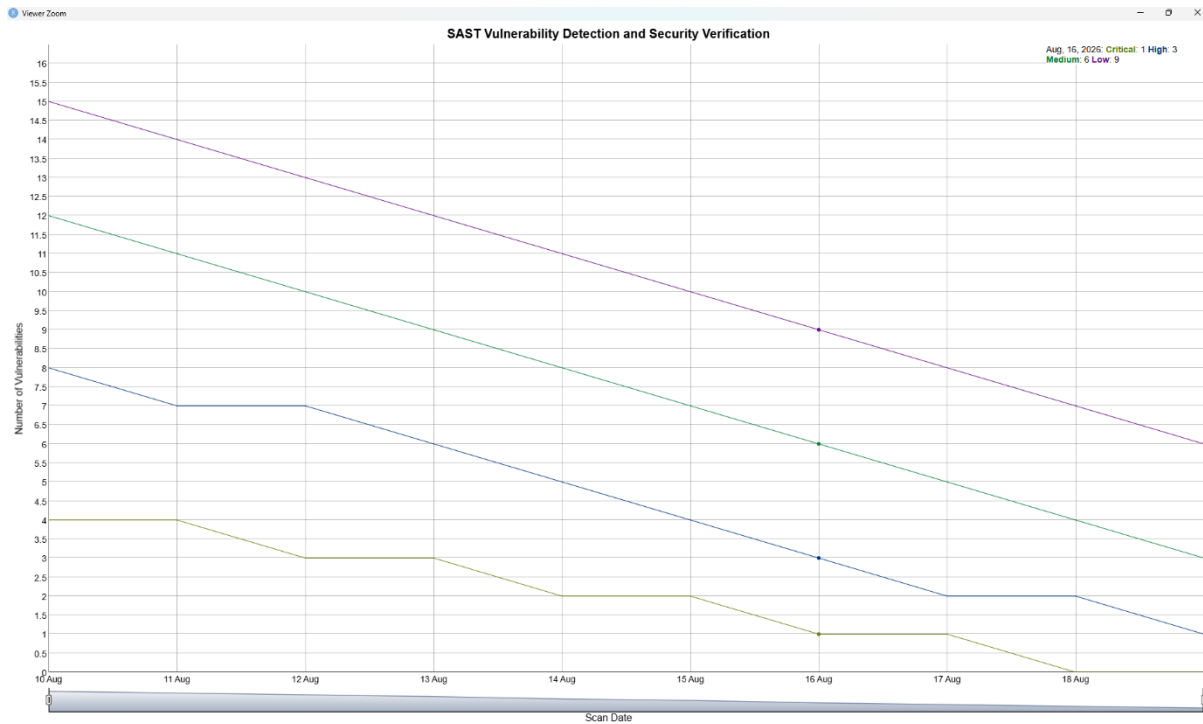
```
sast_data <- xts(  
  cbind(  
    Critical = c(4, 4, 3, 3, 2, 2, 1, 1, 0, 0),  
    High = c(8, 7, 7, 6, 5, 4, 3, 2, 2, 1),  
    Medium = c(12, 11, 10, 9, 8, 7, 6, 5, 4, 3),  
    Low = c(15, 14, 13, 12, 11, 10, 9, 8, 7, 6)  
  ),  
  order.by = days  
)
```

```
# Interactive SAST vulnerability trend  
dygraph(  
  sast_data,
```

```

main = "SAST Vulnerability Detection and Security Verification",
xlab = "Scan Date",
ylab = "Number of Vulnerabilities"
) %>%
dyRangeSelector()

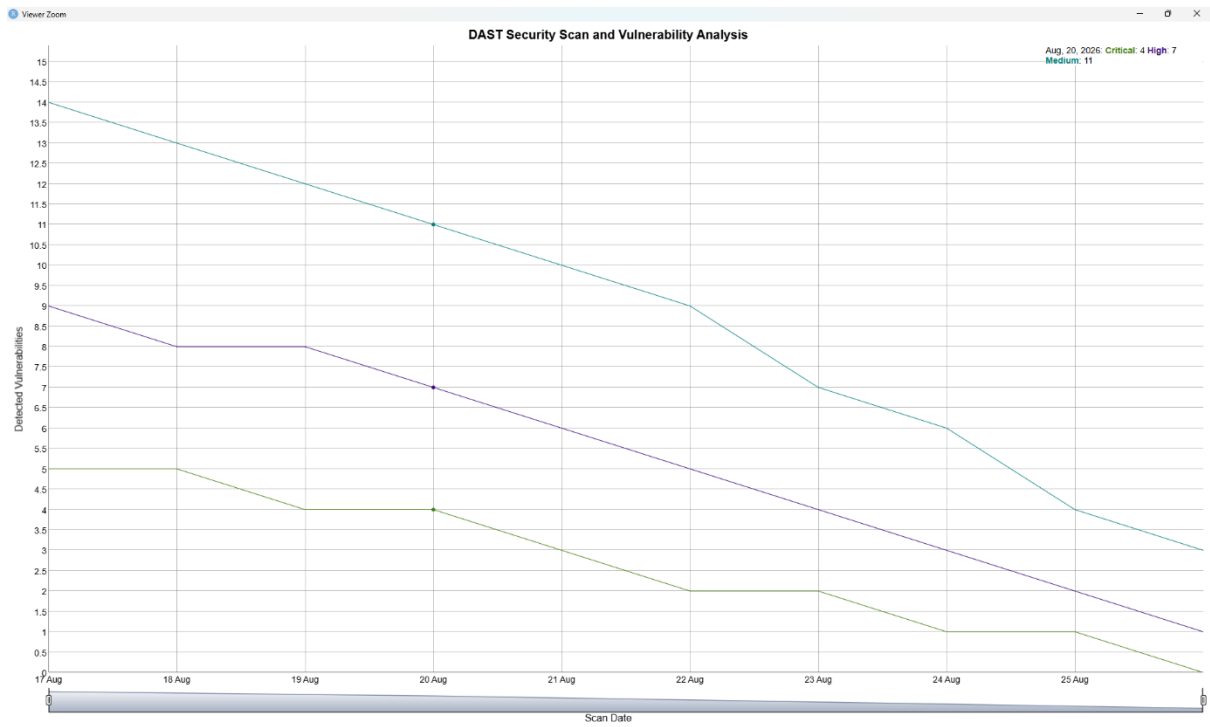
```



```
# =====  
# EXPERIMENT 8  
# Dynamic Application Security Testing (DAST)  
# =====  
  
# Run once if required  
install.packages("dygraphs")  
install.packages("xts")  
  
library(dygraphs)  
library(xts)  
  
# Synthetic DAST scan results  
days <- as.Date("2026-08-17") + 0:9  
  
dast_data <- xts(  
  cbind(  
    Critical = c(5, 5, 4, 4, 3, 2, 2, 1, 1, 0),  
    High = c(9, 8, 8, 7, 6, 5, 4, 3, 2, 1),  
    Medium = c(14, 13, 12, 11, 10, 9, 7, 6, 4, 3)  
  ),  
  order.by = days  
)  
  
dygraph(  
  dast_data,  
  main = "DAST Security Scan and Vulnerability Analysis",  
  xlab = "Scan Date",  
  ylab = "Detected Vulnerabilities"
```

) %>%

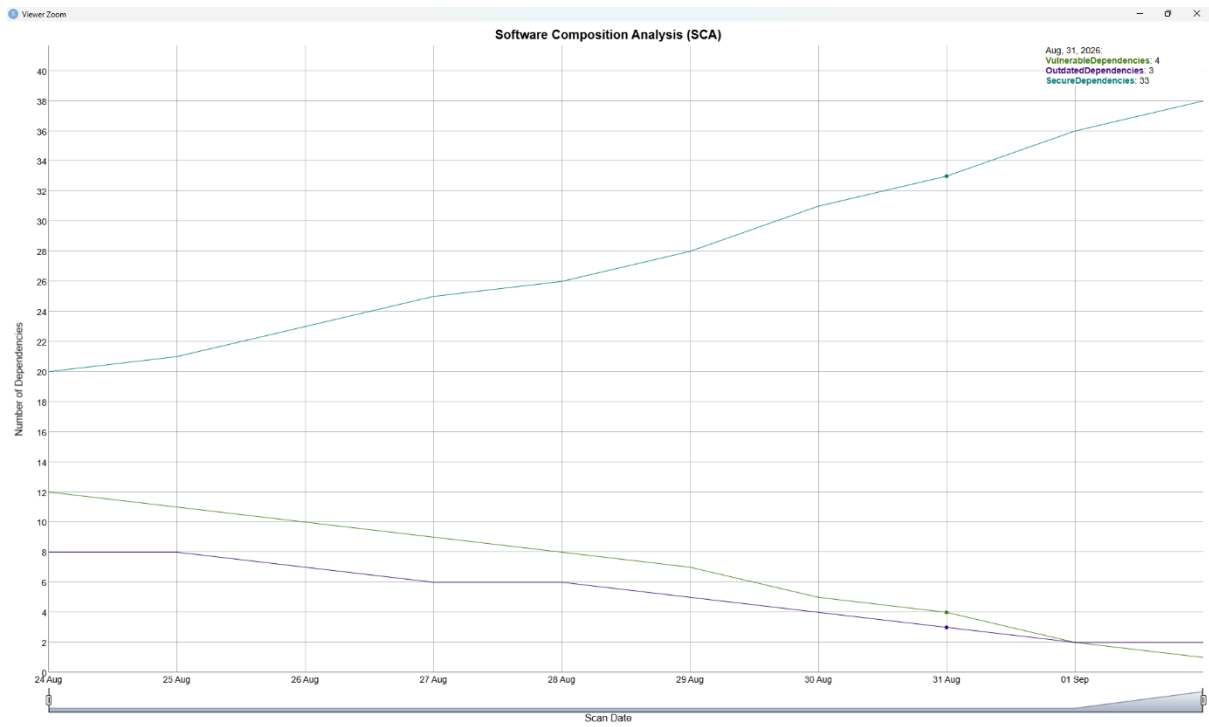
dyRangeSelector()



```
# =====  
# EXPERIMENT 9  
# Software Composition Analysis (SCA)  
# =====  
  
# Run once if required  
install.packages("dygraphs")  
install.packages("xts")  
  
library(dygraphs)  
library(xts)  
  
# Synthetic dependency vulnerability data  
days <- as.Date("2026-08-24") + 0:9  
  
sca_data <- xts(  
  cbind(  
    VulnerableDependencies = c(12, 11, 10, 9, 8, 7, 5, 4, 2, 1),  
    OutdatedDependencies = c(8, 8, 7, 6, 6, 5, 4, 3, 2, 2),  
    SecureDependencies = c(20, 21, 23, 25, 26, 28, 31, 33, 36, 38)  
  ),  
  order.by = days  
)  
  
dygraph(  
  sca_data,  
  main = "Software Composition Analysis (SCA)",  
  xlab = "Scan Date",  
  ylab = "Number of Dependencies"
```

) %>%

dyRangeSelector()

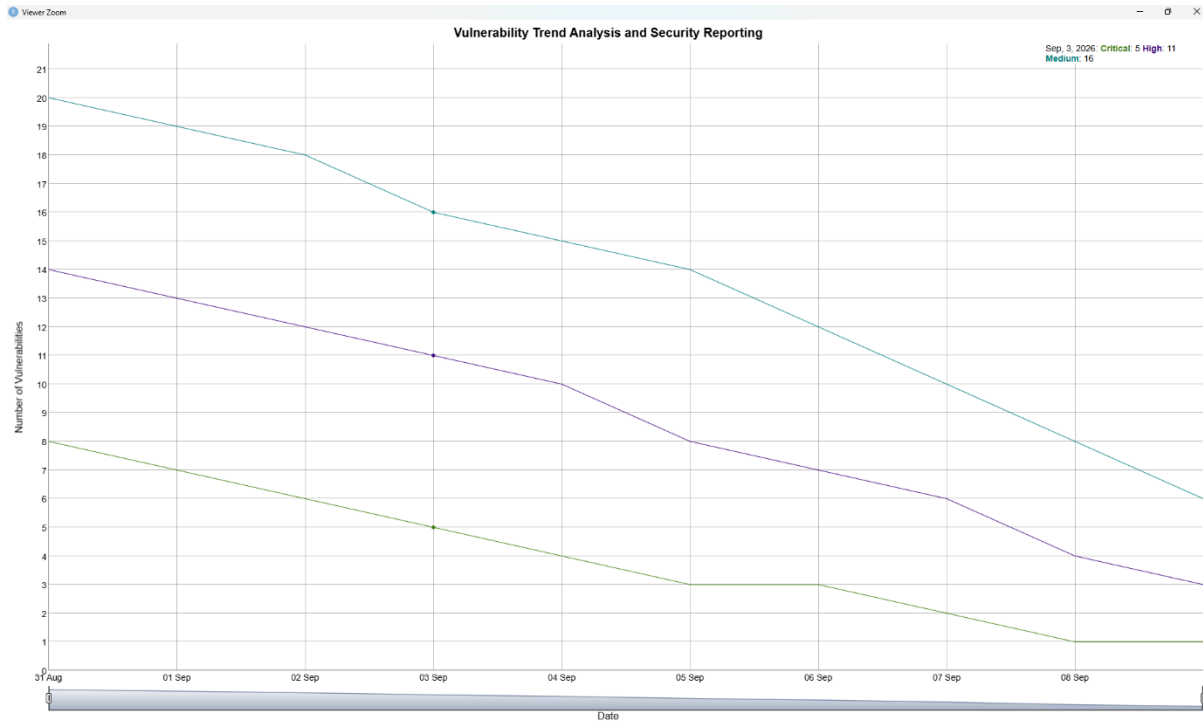


```
# =====  
# EXPERIMENT 10  
# Vulnerability Trend Analysis and Security Reporting  
# =====  
  
# Run once if required  
install.packages("dygraphs")  
install.packages("xts")  
  
library(dygraphs)  
library(xts)  
  
# Synthetic vulnerability trend data  
days <- as.Date("2026-08-31") + 0:9  
  
vulnerability_data <- xts(  
  cbind(  
    Critical = c(8, 7, 6, 5, 4, 3, 3, 2, 1, 1),  
    High = c(14, 13, 12, 11, 10, 8, 7, 6, 4, 3),  
    Medium = c(20, 19, 18, 16, 15, 14, 12, 10, 8, 6)  
  ),  
  order.by = days  
)  
  
# Interactive vulnerability trend report  
dygraph(  
  vulnerability_data,  
  main = "Vulnerability Trend Analysis and Security Reporting",  
  xlab = "Date",
```


ylab = "Number of Vulnerabilities"

) %>%

dyRangeSelector()



```
# =====  
# EXPERIMENT 11  
# Secure Infrastructure Architecture Design using IaC Concepts  
# =====
```

```
# Run once if required  
install.packages("DiagrammeR")
```

```
library(DiagrammeR)
```

```
grViz("  
digraph IaC {
```

```
graph [  
  rankdir=LR,  
  bgcolor=white,  
  label='Secure Infrastructure Architecture using IaC',  
  labelloc=t,  
  fontsize=20  
]
```

```
node [  
  shape=box,  
  style='rounded, filled',  
  fontname=Helvetica,  
  fontsize=13  
]
```

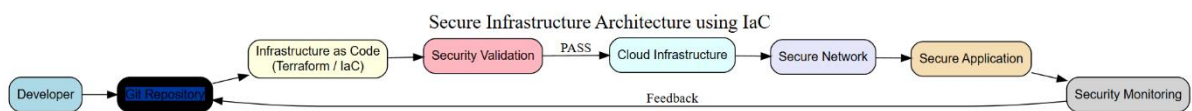
```
Developer [  
  label='Developer',
```

```
    fillcolor=lightblue
]
Git [
    label='Git Repository',
    fillcolor=lightgreen
]
IaC [
    label='Infrastructure as Code\\n(Terraform / IaC)',
    fillcolor=lightyellow
]
Security [
    label='Security Validation',
    fillcolor=lightpink
]
Cloud [
    label='Cloud Infrastructure',
    fillcolor=lightcyan
]
Network [
    label='Secure Network',
    fillcolor=lavender
]
Application [
    label='Secure Application',
    fillcolor=wheat
]
Monitor [
    label='Security Monitoring',
    fillcolor=lightgray
```

```
]
Developer -> Git
Git -> IaC
IaC -> Security
Security -> Cloud [label='PASS']
Cloud -> Network
Network -> Application
Application -> Monitor
Monitor -> Git [label='Feedback']
}
```

```
"
```

Viewer Zoom



```
# =====  
# EXPERIMENT 12  
# Identity and Access Management (IAM) Relationship Modelling  
# =====
```

```
# Run once if required  
install.packages("DiagrammeR")
```

```
library(DiagrammeR)
```

```
grViz("  
digraph IAM {
```

```
graph [  
  rankdir=LR,  
  bgcolor=white,  
  label='Identity and Access Management (IAM) Model',  
  labelloc=t,  
  fontsize=20  
]
```

```
node [  
  shape=box,  
  style='rounded, filled',  
  fontname=Helvetica,  
  fontsize=13  
]
```

```
User [
```

```
    label='Users',  
    fillcolor=lightblue  
]
```

```
Admin [  
    label='Administrator',  
    fillcolor=lightgreen  
]
```

```
Developer [  
    label='Developer',  
    fillcolor=lightyellow  
]
```

```
Role [  
    label='Roles',  
    fillcolor=lightpink  
]
```

```
Permission [  
    label='Permissions',  
    fillcolor=lightcyan  
]
```

```
Resource [  
    label='Resources',  
    fillcolor=lavender  
]
```

```

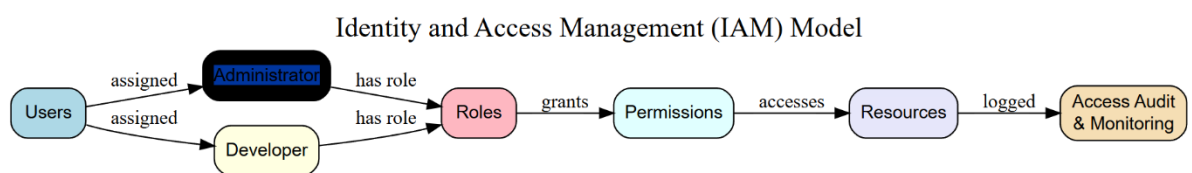
Audit [
  label='Access Audit\\n& Monitoring',
  fillcolor=wheat
]

User -> Admin [label='assigned']
User -> Developer [label='assigned']

Admin -> Role [label='has role']
Developer -> Role [label='has role']
Role -> Permission [label='grants']
Permission -> Resource [label='accesses']
Resource -> Audit [label='logged']
}
")

```

Viewer Zoom



```
# =====  
# EXPERIMENT 13  
# GitOps Workflow Design and Secure Deployment Simulation  
# =====
```

```
# Run once if required  
install.packages("DiagrammeR")
```

```
library(DiagrammeR)
```

```
grViz("  
digraph GitOps {
```

```
graph [  
  rankdir=LR,  
  bgcolor=white,  
  label='GitOps Secure Deployment Workflow',  
  labelloc=t,  
  fontsize=20  
]
```

```
node [  
  shape=box,  
  style='rounded, filled',  
  fontname=Helvetica,  
  fontsize=13  
]
```

```
Developer [  
  shape=oval,  
  style='filled',  
  fontname=Helvetica,  
  fontsize=13  
]
```



```
    label='Developer\\nCode Changes',  
    fillcolor=lightblue  
]
```

```
Git [  
    label='Git Repository\\nVersion Control',  
    fillcolor=lightgreen  
]
```

```
CI [  
    label='CI Pipeline\\nBuild & Test',  
    fillcolor=lightyellow  
]
```

```
Security [  
    label='Security Checks\\nSAST / SCA',  
    fillcolor=lightpink  
]
```

```
CD [  
    label='CD Pipeline\\nDeployment',  
    fillcolor=lightcyan  
]
```

```
Kubernetes [  
    label='Kubernetes\\nCluster',  
    fillcolor=lavender  
]
```

```

Monitor [
  label='Monitoring\\n& Security Verification',
  fillcolor=wheat
]

```

Developer -> Git

Git -> CI

CI -> Security

Security -> CD [label='PASS']

Security -> Git [label='FAIL']

CD -> Kubernetes

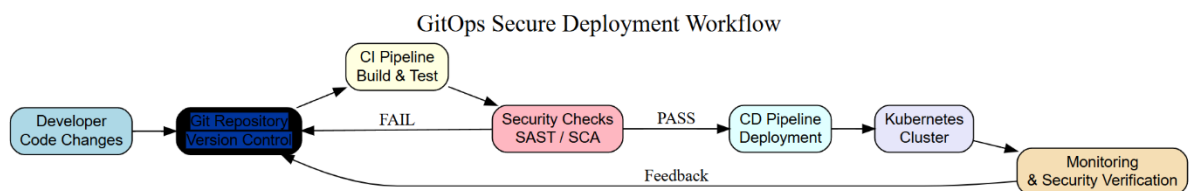
Kubernetes -> Monitor

Monitor -> Git [label='Feedback']

}

")

Viewer Zoom



=====

EXPERIMENT 14

```
# Container Security and Vulnerability Assessment
```

```
# =====
```

```
# Run once if required
```

```
install.packages("dygraphs")
```

```
install.packages("xts")
```

```
library(dygraphs)
```

```
library(xts)
```

```
# Synthetic container security scan data
```

```
days <- as.Date("2026-09-07") + 0:9
```

```
container_data <- xts(
```

```
  cbind(
```

```
    Critical = c(4, 4, 3, 3, 2, 2, 1, 1, 0, 0),
```

```
    High = c(8, 7, 7, 6, 5, 4, 4, 3, 2, 1),
```

```
    Medium = c(12, 11, 10, 9, 8, 7, 6, 5, 4, 3)
```

```
  ),
```

```
  order.by = days
```

```
)
```

```
dygraph(
```

```
  container_data,
```

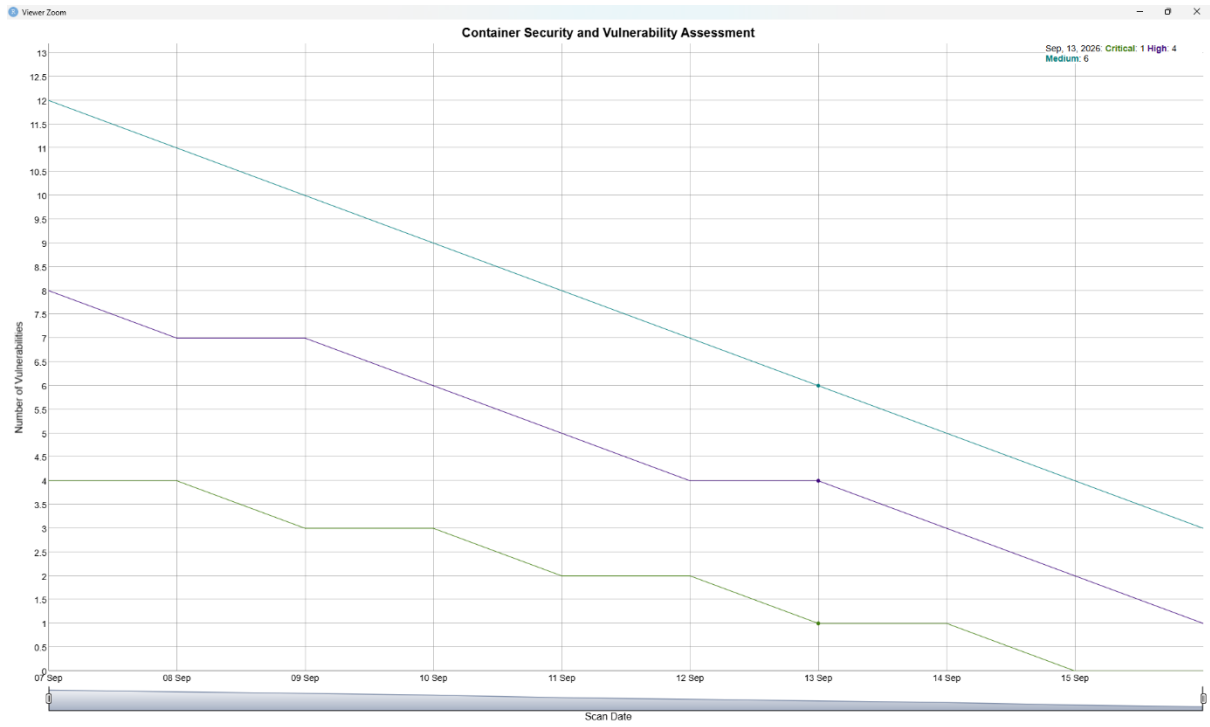
```
  main = "Container Security and Vulnerability Assessment",
```

```
  xlab = "Scan Date",
```

```
  ylab = "Number of Vulnerabilities"
```

```
) %>%
```

```
  dyRangeSelector()
```

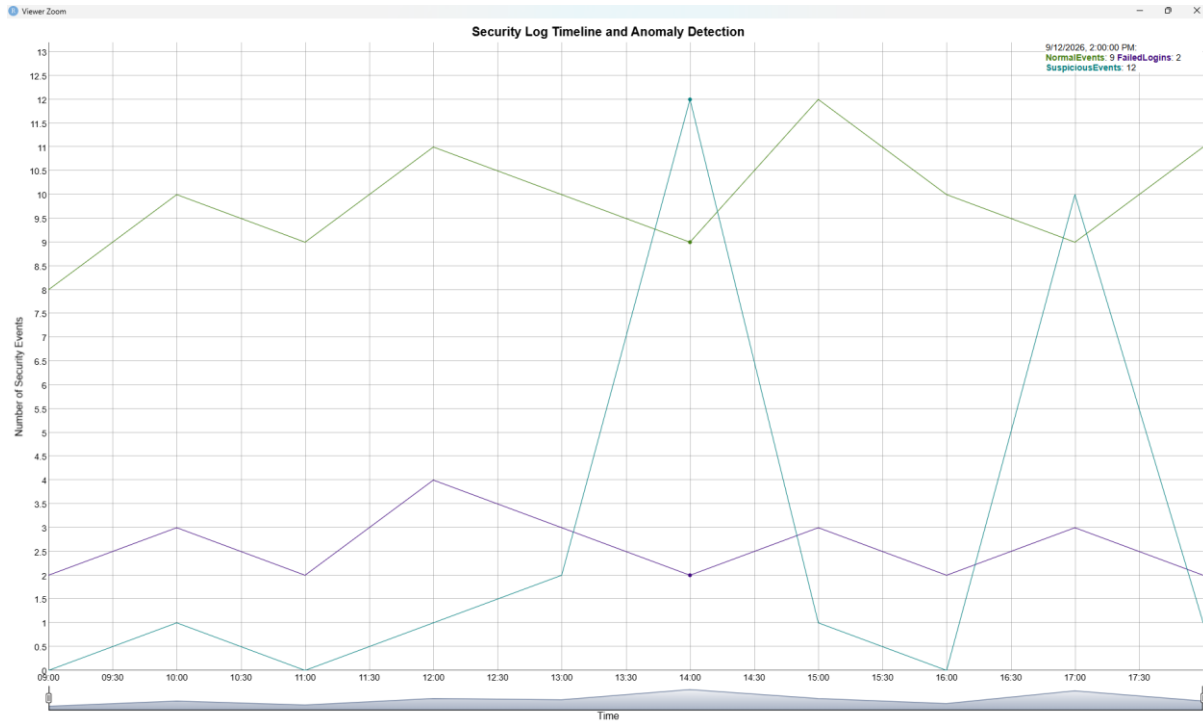


```
# =====  
# EXPERIMENT 15  
# Security Log Generation, Incident Timeline Analysis  
# and Anomaly Detection  
# =====  
  
# Run once if required  
install.packages("dygraphs")  
install.packages("xts")  
  
library(dygraphs)  
library(xts)  
  
# Synthetic security log data  
time <- as.POSIXct("2026-09-12 09:00:00") + (0:9) * 3600  
  
log_data <- xts(  
  cbind(  
    NormalEvents = c(8, 10, 9, 11, 10, 9, 12, 10, 9, 11),  
    FailedLogins = c(2, 3, 2, 4, 3, 2, 3, 2, 3, 2),  
    SuspiciousEvents = c(0, 1, 0, 1, 2, 12, 1, 0, 10, 1)  
  ),  
  order.by = time  
)  
  
# Interactive security log and anomaly analysis  
dygraph(  
  log_data,  
  main = "Security Log Timeline and Anomaly Detection",
```

```

xlab = "Time",
ylab = "Number of Security Events"
) %>%
dyRangeSelector()

```



```
# =====  
# EXPERIMENT 16  
# Development of a DevSecOps Security Monitoring Dashboard  
# =====
```

```
# Run once if required
```

```
install.packages("dygraphs")
```

```
install.packages("xts")
```

```
library(dygraphs)
```

```
library(xts)
```

```
# Synthetic security monitoring data
```

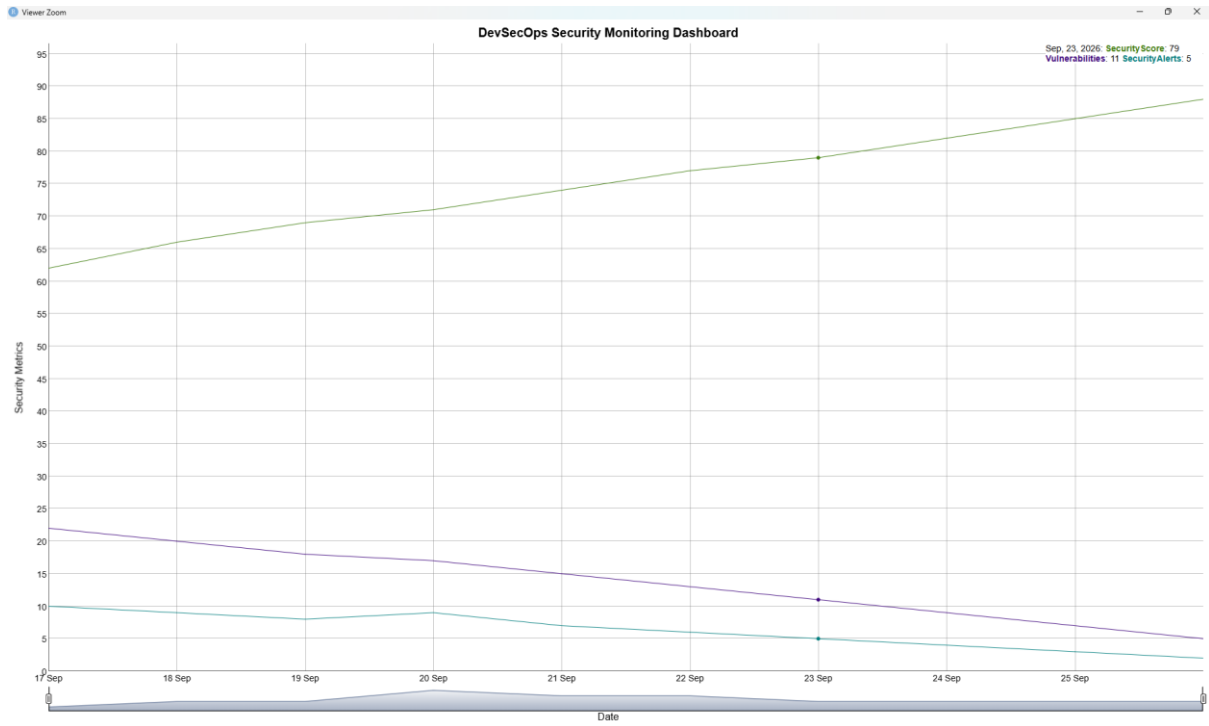
```
days <- as.Date("2026-09-17") + 0:9
```

```
dashboard_data <- xts(  
  cbind(  
    SecurityScore = c(62, 66, 69, 71, 74, 77, 79, 82, 85, 88),  
    Vulnerabilities = c(22, 20, 18, 17, 15, 13, 11, 9, 7, 5),  
    SecurityAlerts = c(10, 9, 8, 9, 7, 6, 5, 4, 3, 2)  
  ),  
  order.by = days  
)
```

```
dygraph(  
  dashboard_data,  
  main = "DevSecOps Security Monitoring Dashboard",  
  xlab = "Date",  
  ylab = "Security Metrics"
```

) %>%

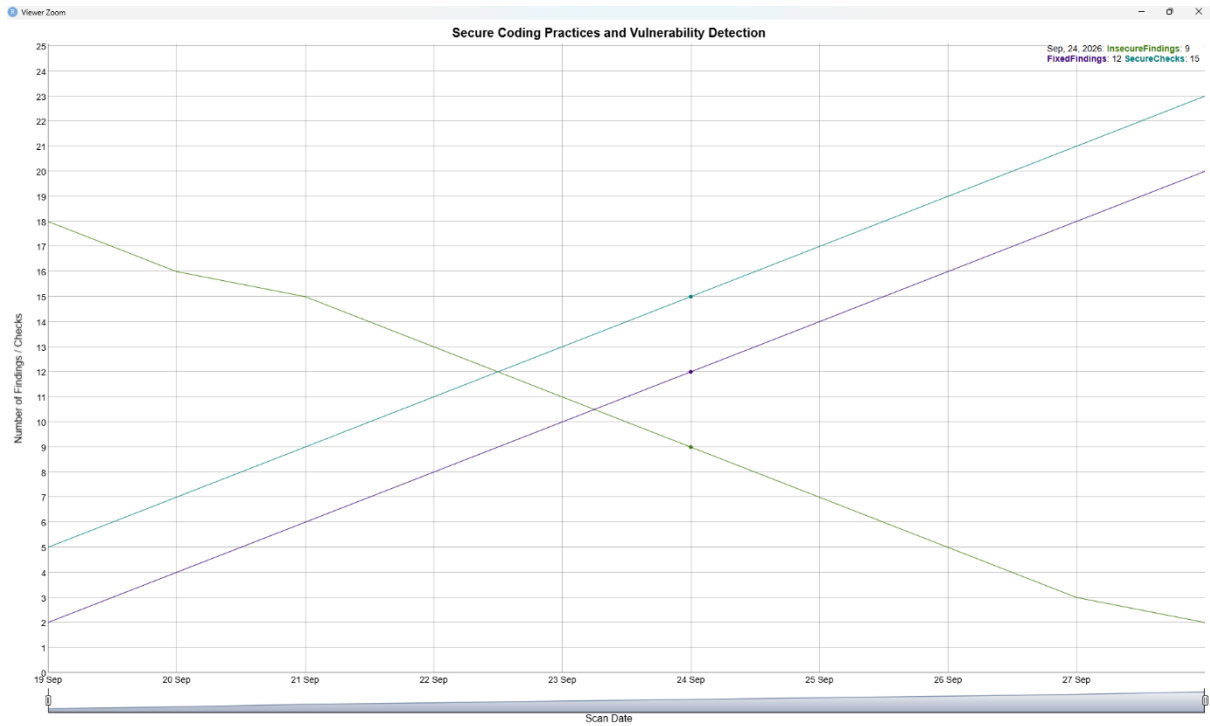
dyRangeSelector()




```
# =====  
# EXPERIMENT 17  
# Secure Coding Practices and Vulnerability Detection Simulation  
# =====  
  
# Run once if required  
install.packages("dygraphs")  
install.packages("xts")  
  
library(dygraphs)  
library(xts)  
  
# Synthetic secure coding scan data  
days <- as.Date("2026-09-19") + 0:9  
  
coding_data <- xts(  
  cbind(  
    InsecureFindings = c(18, 16, 15, 13, 11, 9, 7, 5, 3, 2),  
    FixedFindings = c(2, 4, 6, 8, 10, 12, 14, 16, 18, 20),  
    SecureChecks = c(5, 7, 9, 11, 13, 15, 17, 19, 21, 23)  
  ),  
  order.by = days  
)  
  
dygraph(  
  coding_data,  
  main = "Secure Coding Practices and Vulnerability Detection",  
  xlab = "Scan Date",  
  ylab = "Number of Findings / Checks"
```

) %>%

dyRangeSelector()



```
# =====  
# EXPERIMENT 18  
# OWASP Top 10 Vulnerability Identification and Risk Classification  
# =====
```

```
# Run once if required  
install.packages("DiagrammeR")
```

```
library(DiagrammeR)
```

```
grViz("  
digraph OWASP {
```

```
graph [  
  rankdir=LR,  
  bgcolor=white,  
  label='OWASP Top 10 Vulnerability Identification and Risk Classification',  
  labelloc=t,  
  fontsize=20  
]
```

```
node [  
  shape=box,  
  style='rounded, filled',  
  fontname=Helvetica,  
  fontsize=12  
]
```

```
App [  
  graph TD  
    subgraph OWASP_Top_10_Vulnerabilities [OWASP Top 10 Vulnerabilities]  
      direction LR  
      A[A1] --- B[B1] --- C[C1] --- D[D1] --- E[E1] --- F[F1] --- G[G1] --- H[H1] --- I[I1] --- J[J1]  
    end  
    A --- B --- C --- D --- E --- F --- G --- H --- I --- J  
  ]
```

```
    label='Web Application',  
    fillcolor=lightblue  
]
```

```
A01 [  
    label='A01 Broken Access Control',  
    fillcolor=lightpink  
]
```

```
A02 [  
    label='A02 Cryptographic Failures',  
    fillcolor=lightyellow  
]
```

```
A03 [  
    label='A03 Injection',  
    fillcolor=lightcyan  
]
```

```
A04 [  
    label='A04 Insecure Design',  
    fillcolor=lavender  
]
```

```
A05 [  
    label='A05 Security Misconfiguration',  
    fillcolor=wheat  
]
```

```
A06 [  
    label='A06 Vulnerable Components',  
    fillcolor=lightgreen  
]
```

```
Risk [  
    label='Risk Classification\\nCritical / High / Medium / Low',  
    shape=diamond,  
    fillcolor=lightgray  
]
```

App -> A01

App -> A02

App -> A03

App -> A04

App -> A05

App -> A06

A01 -> Risk

A02 -> Risk

A03 -> Risk

A04 -> Risk

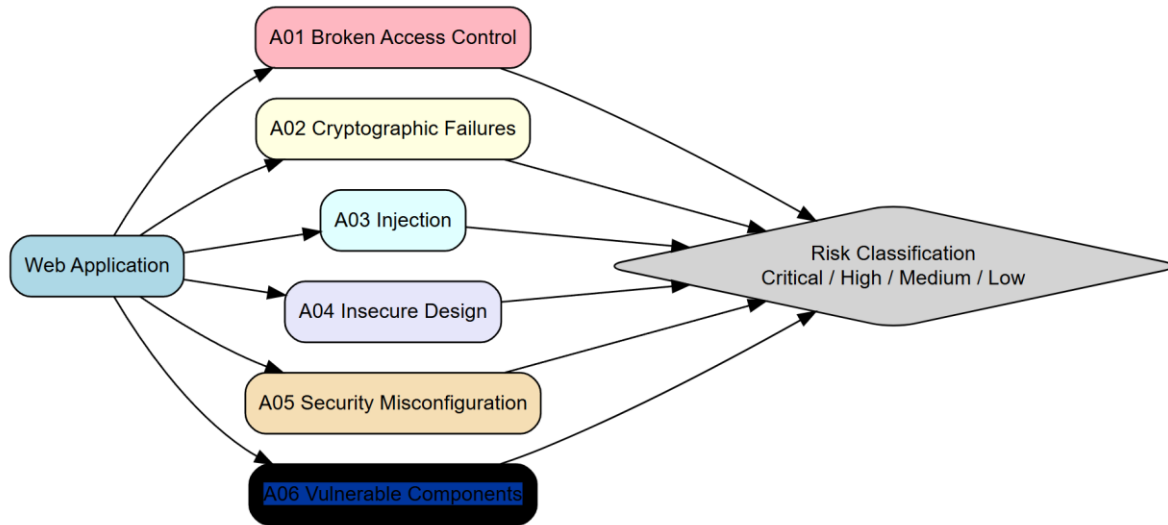
A05 -> Risk

A06 -> Risk

}

")

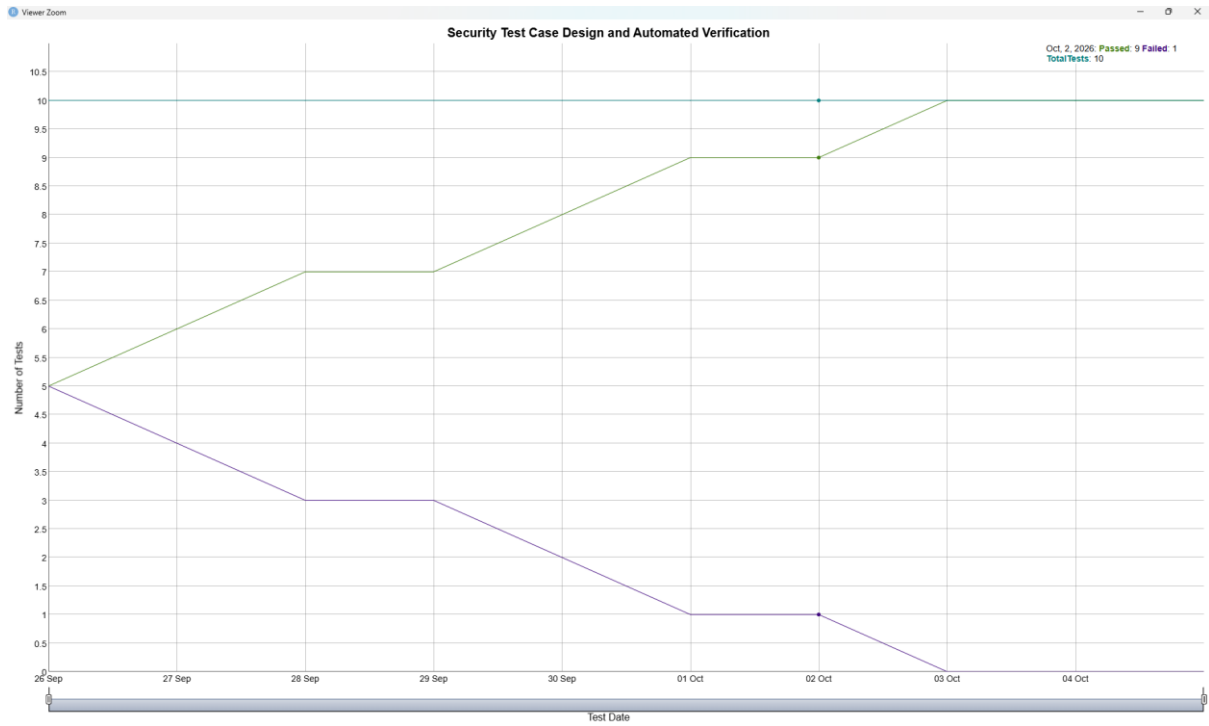
OWASP Top 10 Vulnerability Identification and Risk Classification



```
# =====  
# EXPERIMENT 19  
# Security Test Case Design and Automated Verification Simulation  
# =====  
  
# Run once if required  
install.packages("dygraphs")  
install.packages("xts")  
  
library(dygraphs)  
library(xts)  
  
# Synthetic automated security test results  
days <- as.Date("2026-09-26") + 0:9  
  
test_data <- xts(  
  cbind(  
    Passed = c(5, 6, 7, 7, 8, 9, 9, 10, 10, 10),  
    Failed = c(5, 4, 3, 3, 2, 1, 1, 0, 0, 0),  
    TotalTests = c(10, 10, 10, 10, 10, 10, 10, 10, 10, 10)  
  ),  
  order.by = days  
)  
  
dygraph(  
  test_data,  
  main = "Security Test Case Design and Automated Verification",  
  xlab = "Test Date",  
  ylab = "Number of Tests"
```

) %>%

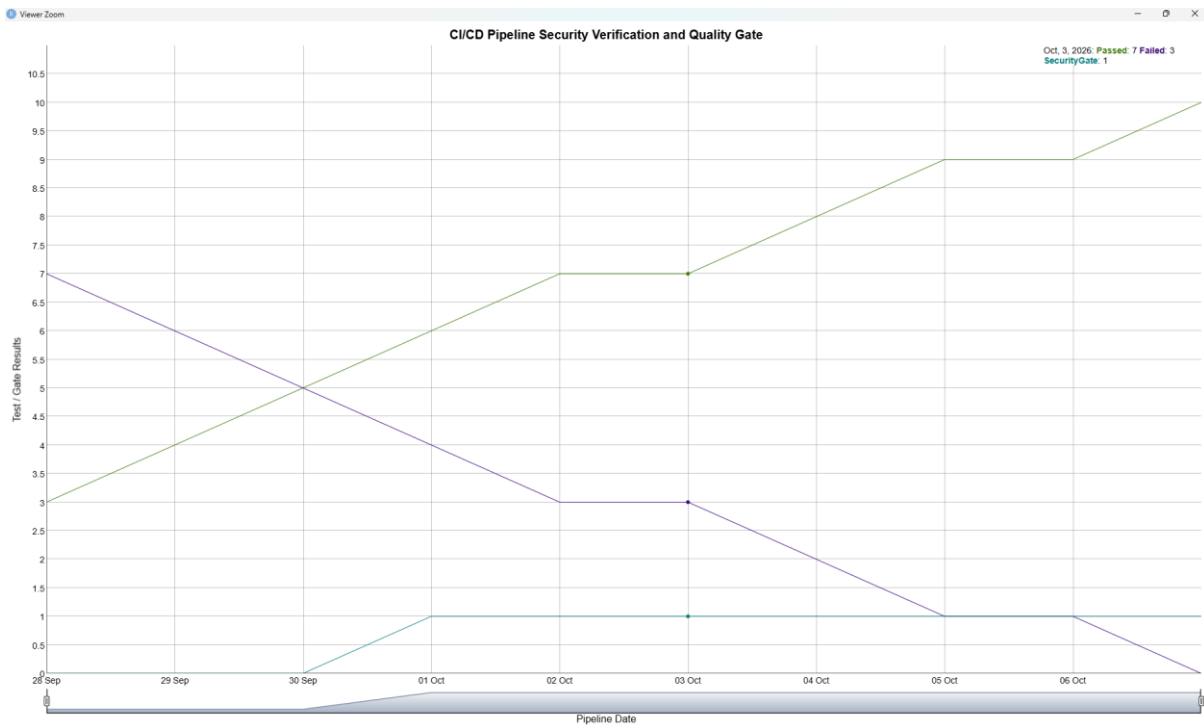
dyRangeSelector()




```
# =====  
# EXPERIMENT 20  
# CI/CD Pipeline Security Verification and Quality Gate Simulation  
# =====  
  
# Run once if required  
install.packages("dygraphs")  
install.packages("xts")  
  
library(dygraphs)  
library(xts)  
  
# Synthetic CI/CD security verification data  
days <- as.Date("2026-09-28") + 0:9  
  
pipeline_data <- xts(  
  cbind(  
    Passed = c(3, 4, 5, 6, 7, 7, 8, 9, 9, 10),  
    Failed = c(7, 6, 5, 4, 3, 3, 2, 1, 1, 0),  
    SecurityGate = c(0, 0, 0, 1, 1, 1, 1, 1, 1, 1)  
  ),  
  order.by = days  
)  
  
dygraph(  
  pipeline_data,  
  main = "CI/CD Pipeline Security Verification and Quality Gate",  
  xlab = "Pipeline Date",  
  ylab = "Test / Gate Results"
```

) %>%

dyRangeSelector()



```

# =====
# EXPERIMENT 21
# Security Vulnerability Prioritization using CVSS-Based Scoring
# =====

# Run once if required
install.packages("dygraphs")
install.packages("xts")

library(dygraphs)
library(xts)

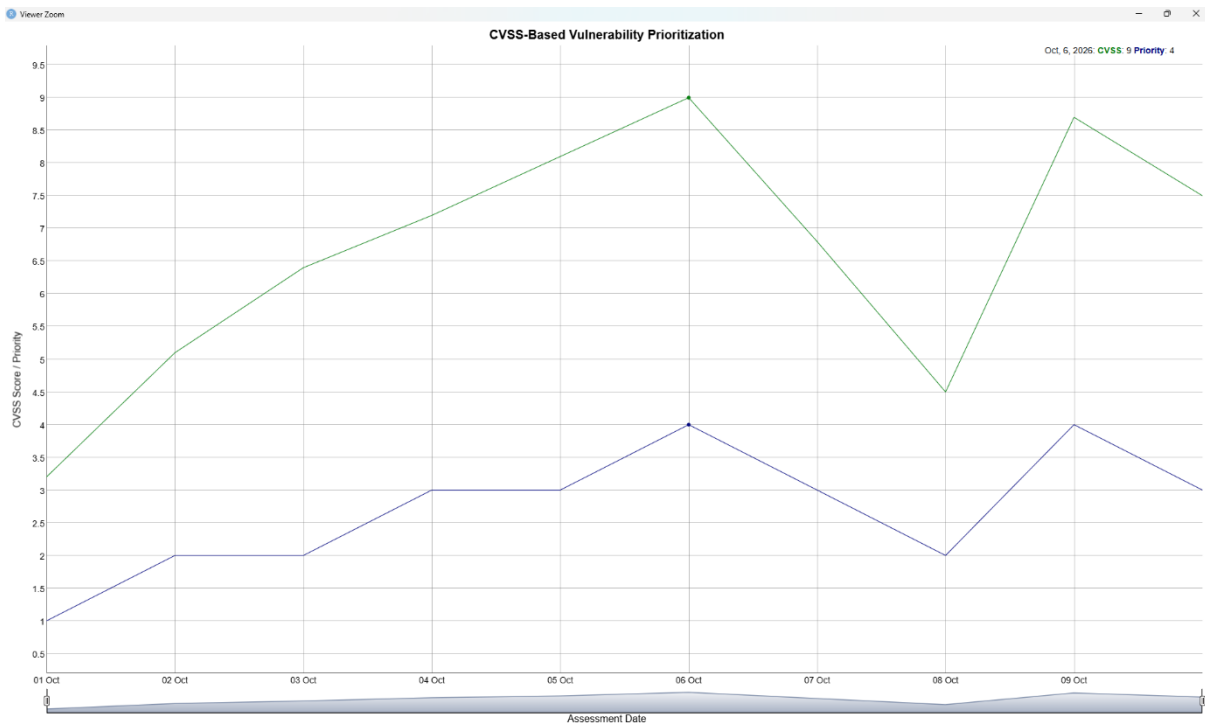
# Synthetic CVSS vulnerability scores
days <- as.Date("2026-10-01") + 0:9

cvss_data <- xts(
  cbind(
    CVSS = c(3.2, 5.1, 6.4, 7.2, 8.1, 9.0, 6.8, 4.5, 8.7, 7.5),
    Priority = c(1, 2, 2, 3, 3, 4, 3, 2, 4, 3)
  ),
  order.by = days
)

dygraph(
  cvss_data,
  main = "CVSS-Based Vulnerability Prioritization",
  xlab = "Assessment Date",
  ylab = "CVSS Score / Priority"
) %>%

```

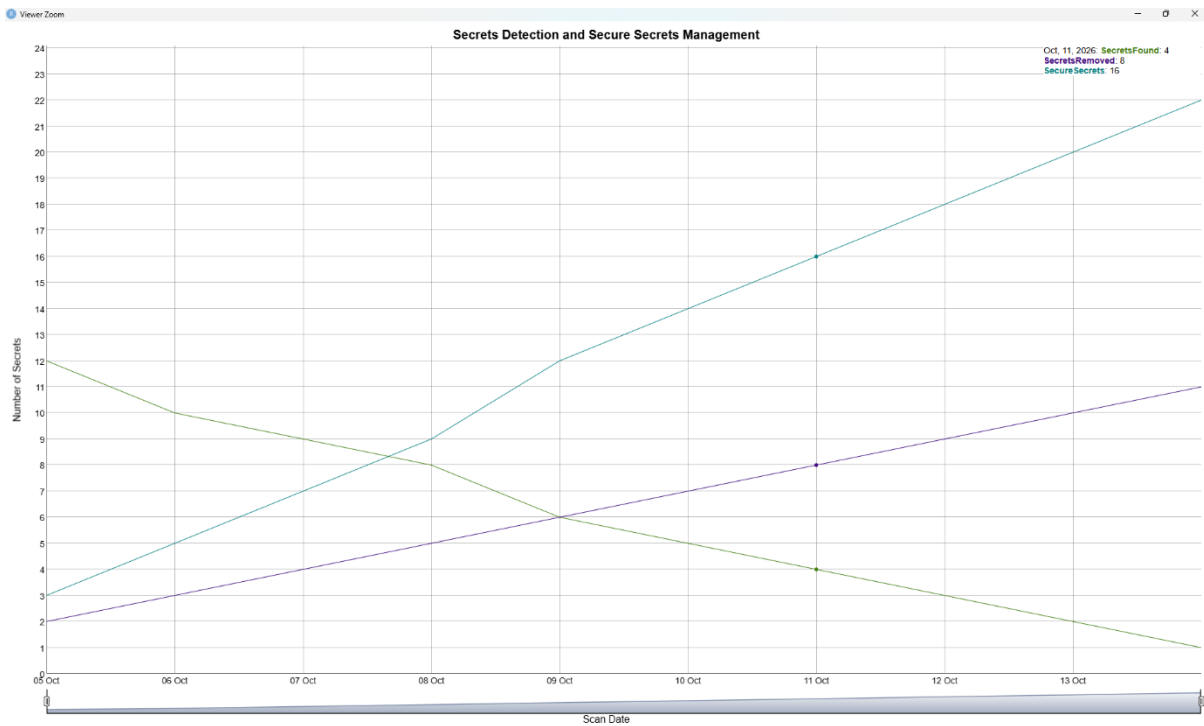
dyRangeSelector()



```
# =====  
# EXPERIMENT 22  
# Secrets Detection and Secure Secrets Management Simulation  
# =====  
  
# Run once if required  
install.packages("dygraphs")  
install.packages("xts")  
  
library(dygraphs)  
library(xts)  
  
# Synthetic secrets detection data  
days <- as.Date("2026-10-05") + 0:9  
  
secrets_data <- xts(  
  cbind(  
    SecretsFound = c(12, 10, 9, 8, 6, 5, 4, 3, 2, 1),  
    SecretsRemoved = c(2, 3, 4, 5, 6, 7, 8, 9, 10, 11),  
    SecureSecrets = c(3, 5, 7, 9, 12, 14, 16, 18, 20, 22)  
  ),  
  order.by = days  
)  
  
dygraph(  
  secrets_data,  
  main = "Secrets Detection and Secure Secrets Management",  
  xlab = "Scan Date",  
  ylab = "Number of Secrets"
```

) %>%

dyRangeSelector()



```
# =====  
# EXPERIMENT 23  
# API Security Testing and Vulnerability Analysis  
# =====
```

```
# Run once if required
```

```
install.packages("dygraphs")
```

```
install.packages("xts")
```

```
library(dygraphs)
```

```
library(xts)
```

```
# Synthetic API security testing data
```

```
days <- as.Date("2026-10-07") + 0:9
```

```
api_data <- xts(
```

```
  cbind(
```

```
    Critical = c(3, 3, 2, 2, 1, 1, 1, 0, 0, 0),
```

```
    High = c(7, 6, 6, 5, 4, 4, 3, 2, 1, 1),
```

```
    Medium = c(10, 9, 8, 8, 7, 6, 5, 4, 3, 2),
```

```
    SecurityTests = c(20, 22, 24, 25, 27, 29, 31, 33, 35, 37)
```

```
  ),
```

```
  order.by = days
```

```
)
```

```
dygraph(
```

```
  api_data,
```

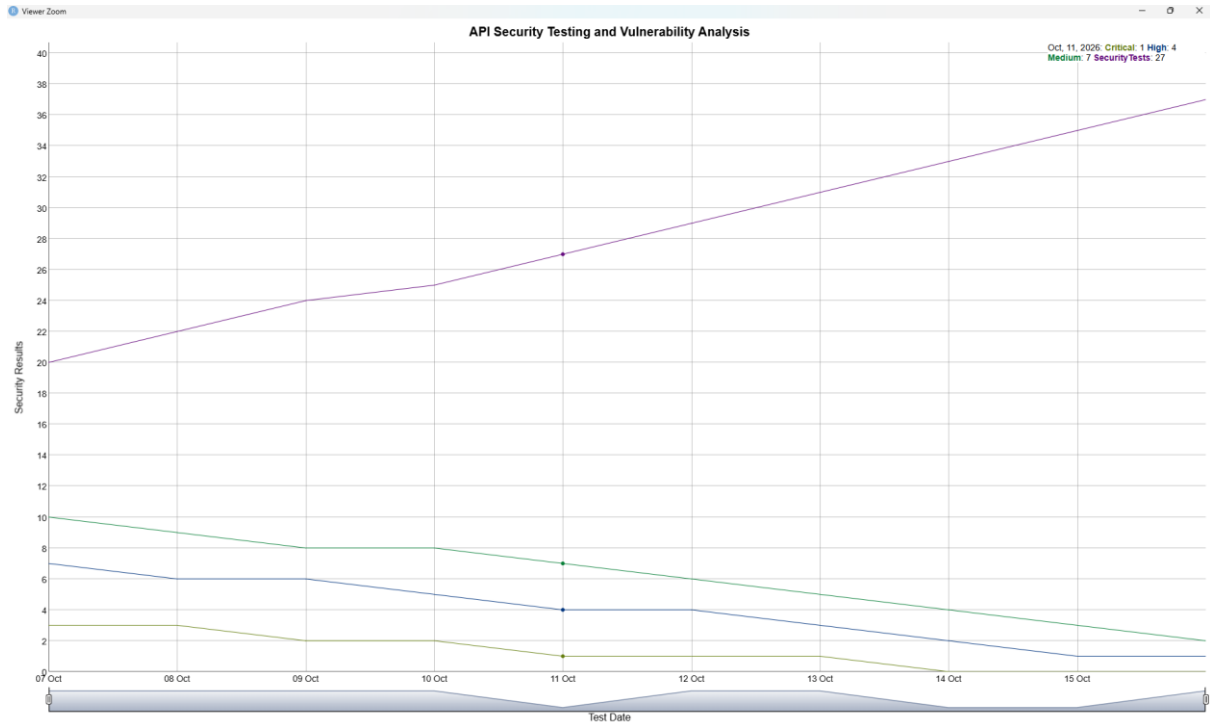
```
  main = "API Security Testing and Vulnerability Analysis",
```

```
  xlab = "Test Date",
```

```
ylab = "Security Results"
```

```
) %>%
```

```
dyRangeSelector()
```

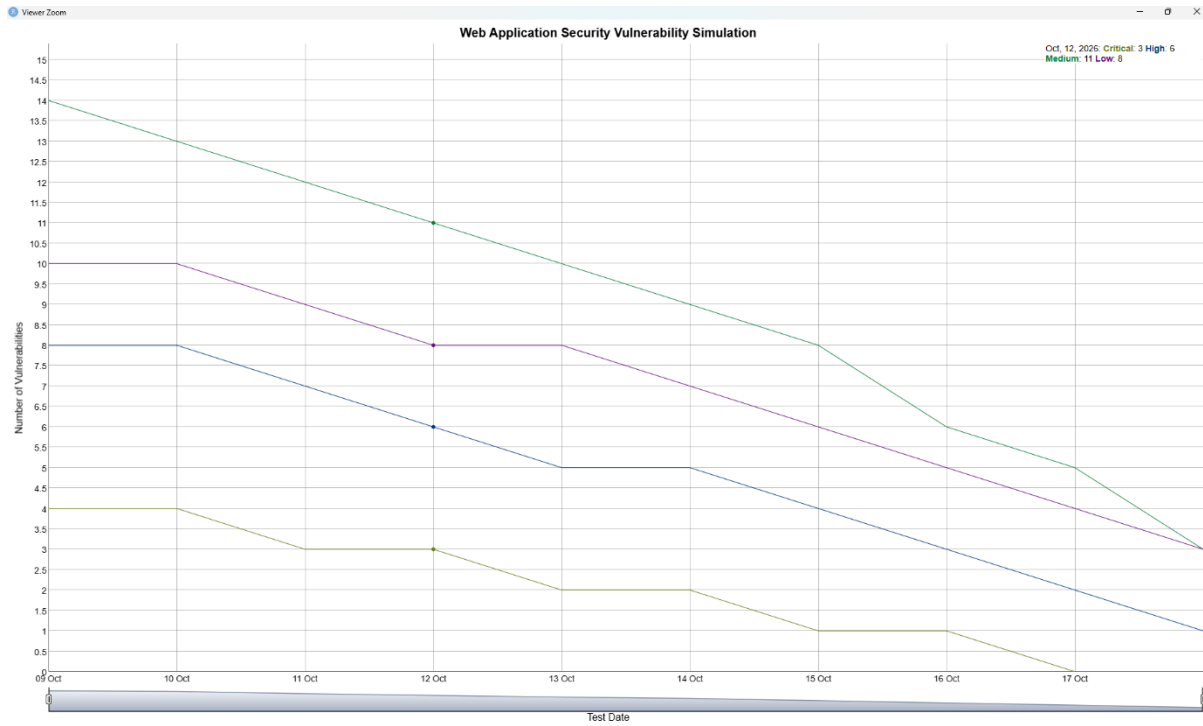



```
# =====  
# EXPERIMENT 24  
# Web Application Security Vulnerability Simulation  
# =====  
  
# Run once if required  
install.packages("dygraphs")  
install.packages("xts")  
  
library(dygraphs)  
library(xts)  
  
# Synthetic web application vulnerability data  
days <- as.Date("2026-10-09") + 0:9  
  
web_data <- xts(  
  cbind(  
    Critical = c(4, 4, 3, 3, 2, 2, 1, 1, 0, 0),  
    High = c(8, 8, 7, 6, 5, 5, 4, 3, 2, 1),  
    Medium = c(14, 13, 12, 11, 10, 9, 8, 6, 5, 3),  
    Low = c(10, 10, 9, 8, 8, 7, 6, 5, 4, 3)  
  ),  
  order.by = days  
)  
  
dygraph(  
  web_data,  
  main = "Web Application Security Vulnerability Simulation",  
  xlab = "Test Date",
```

```
ylab = "Number of Vulnerabilities"
```

```
) %>%
```

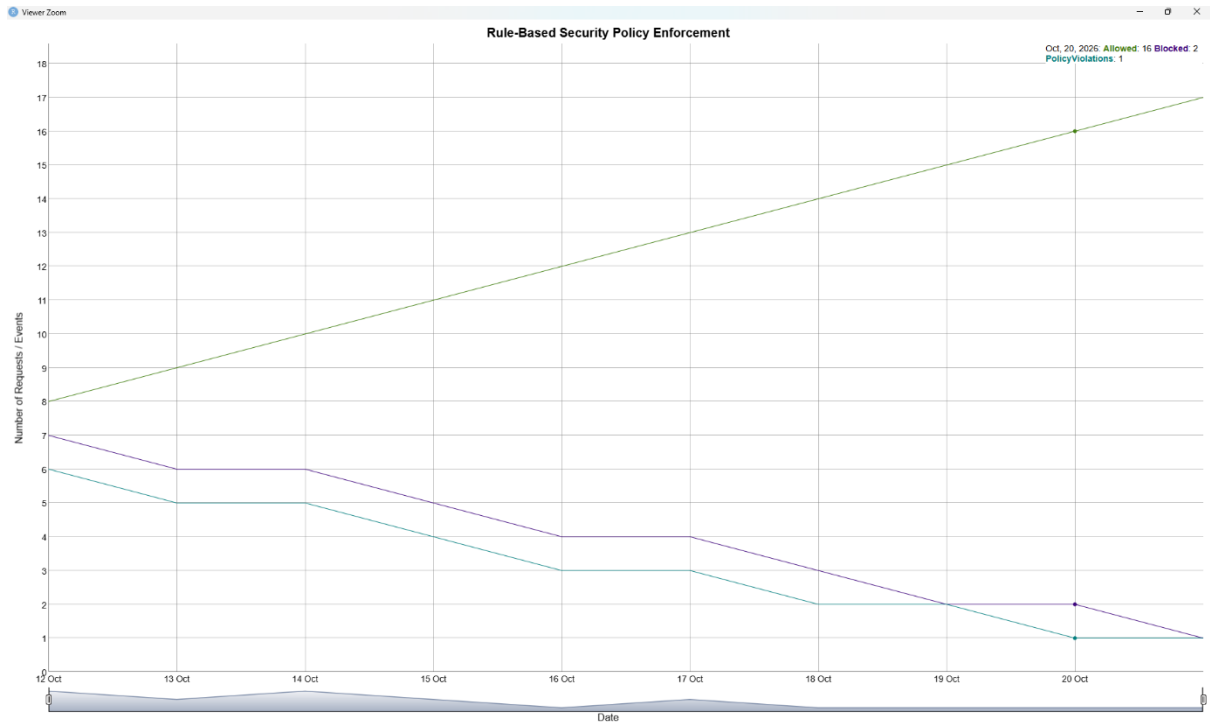
```
dyRangeSelector()
```



```
# =====  
# EXPERIMENT 25  
# Security Policy Enforcement using Rule-Based Simulation  
# =====  
  
# Run once if required  
install.packages("dygraphs")  
install.packages("xts")  
  
library(dygraphs)  
library(xts)  
  
# Synthetic security policy enforcement data  
days <- as.Date("2026-10-12") + 0:9  
  
policy_data <- xts(  
  cbind(  
    Allowed = c(8, 9, 10, 11, 12, 13, 14, 15, 16, 17),  
    Blocked = c(7, 6, 6, 5, 4, 4, 3, 2, 2, 1),  
    PolicyViolations = c(6, 5, 5, 4, 3, 3, 2, 2, 1, 1)  
  ),  
  order.by = days  
)  
  
dygraph(  
  policy_data,  
  main = "Rule-Based Security Policy Enforcement",  
  xlab = "Date",  
  ylab = "Number of Requests / Events"
```

) %>%

dyRangeSelector()



```
# =====  
# EXPERIMENT 26  
# DevSecOps Security Gate Modelling and Decision Simulation  
# =====
```

```
# Run once if required  
install.packages("DiagrammeR")
```

```
library(DiagrammeR)
```

```
grViz("  
digraph SecurityGate {
```

```
graph [  
  rankdir=LR,  
  bgcolor=white,  
  label='DevSecOps Security Gate Decision Workflow',  
  labelloc=t,  
  fontsize=20  
]
```

```
node [  
  shape=box,  
  style='rounded, filled',  
  fontname=Helvetica,  
  fontsize=13  
]
```

```
Start [
```

```
    label='Start',  
    fillcolor=lightblue  
]
```

```
Build [  
    label='Application Build',  
    fillcolor=lightgreen  
]
```

```
SecurityScan [  
    label='Security Scan',  
    fillcolor=lightyellow  
]
```

```
Gate [  
    label='Security Gate\\nDecision',  
    shape=diamond,  
    fillcolor=lightpink  
]
```

```
Deploy [  
    label='Deploy Application',  
    fillcolor=lightcyan  
]
```

```
Reject [  
    label='Reject Build\\n& Fix Issues',  
    fillcolor=wheat  
]
```

Start -> Build

Build -> SecurityScan

SecurityScan -> Gate

Gate -> Deploy [label='PASS']

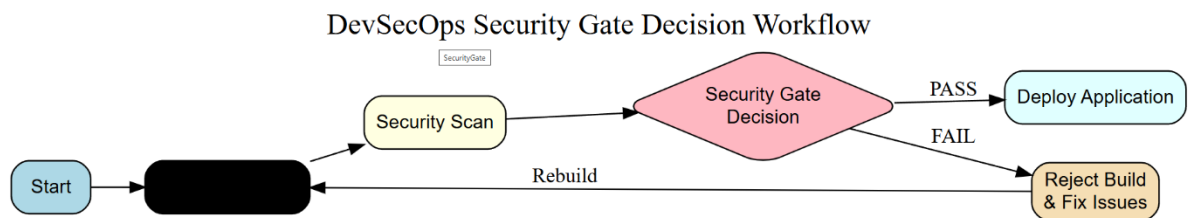
Gate -> Reject [label='FAIL']

Reject -> Build [label='Rebuild']

}

")

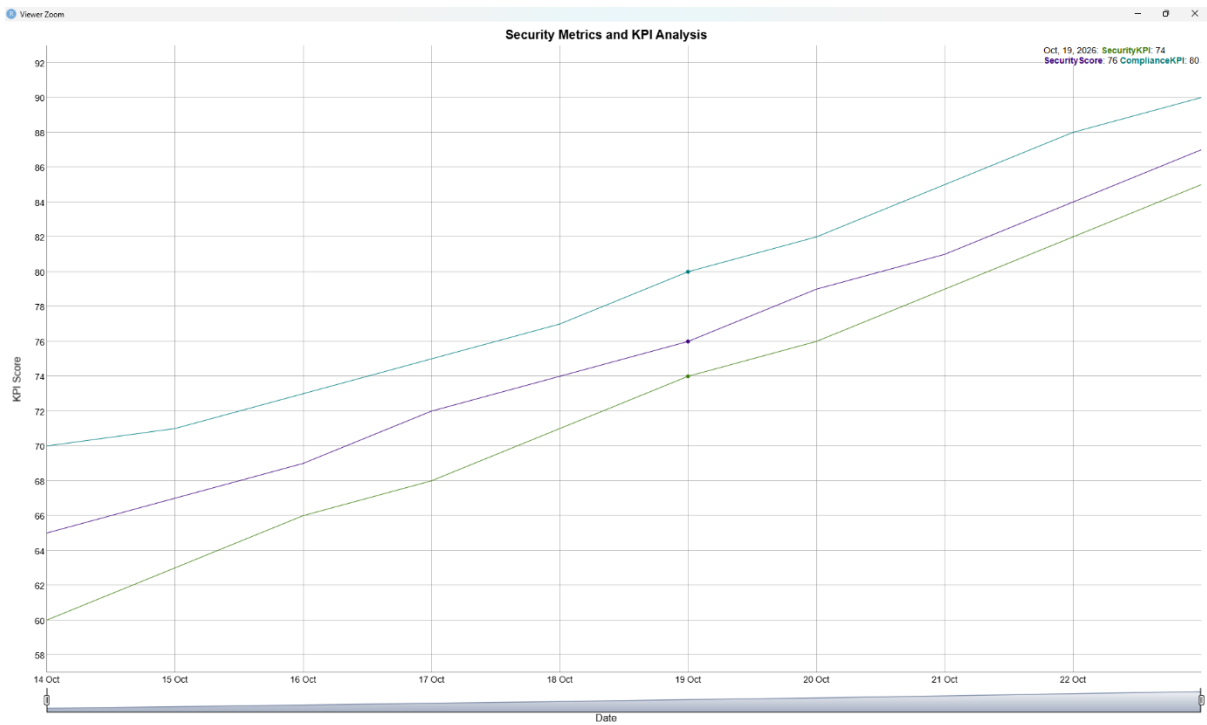
Viewer Zoom



```
# =====  
# EXPERIMENT 27  
# Security Metrics Generation and KPI Analysis  
# =====  
  
# Run once if required  
install.packages("dygraphs")  
install.packages("xts")  
  
library(dygraphs)  
library(xts)  
  
# Synthetic security KPI data  
days <- as.Date("2026-10-14") + 0:9  
  
kpi_data <- xts(  
  cbind(  
    SecurityKPI = c(60, 63, 66, 68, 71, 74, 76, 79, 82, 85),  
    SecurityScore = c(65, 67, 69, 72, 74, 76, 79, 81, 84, 87),  
    ComplianceKPI = c(70, 71, 73, 75, 77, 80, 82, 85, 88, 90)  
  ),  
  order.by = days  
)  
  
dygraph(  
  kpi_data,  
  main = "Security Metrics and KPI Analysis",  
  xlab = "Date",  
  ylab = "KPI Score"
```


) %>%

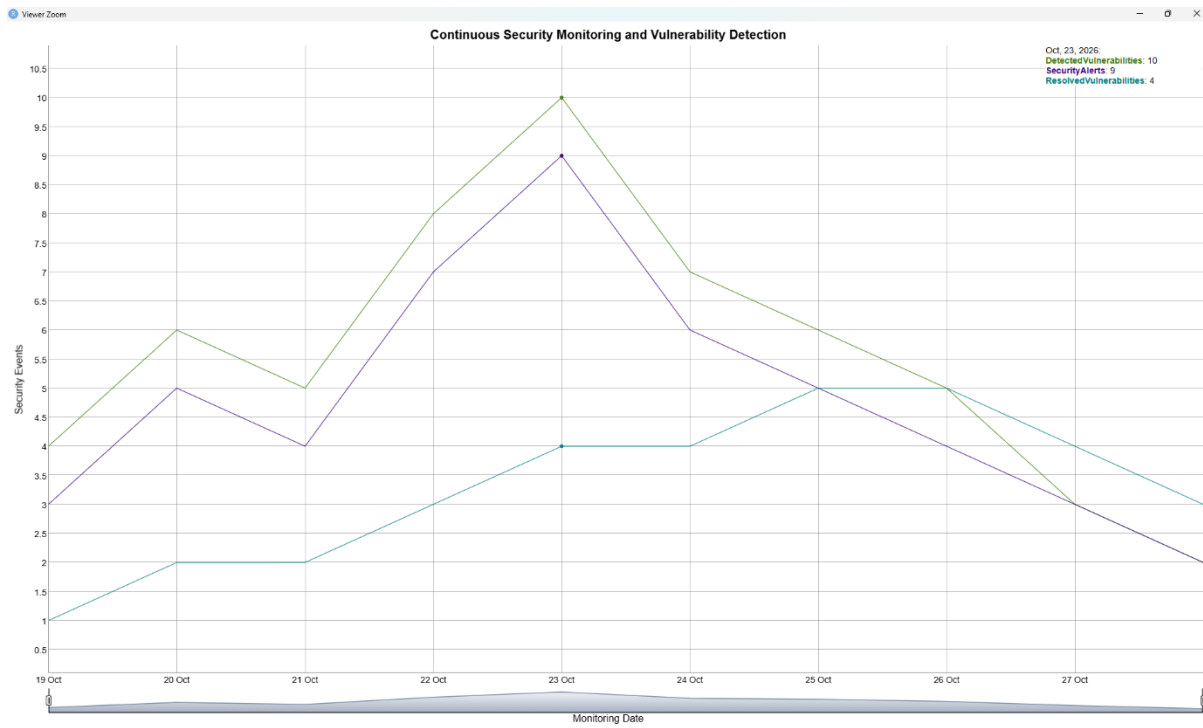
dyRangeSelector()



```
# =====  
# EXPERIMENT 28  
# Continuous Security Monitoring and Vulnerability Detection Simulation  
# =====  
  
# Run once if required  
install.packages("dygraphs")  
install.packages("xts")  
  
library(dygraphs)  
library(xts)  
  
# Synthetic continuous security monitoring data  
days <- as.Date("2026-10-19") + 0:9  
  
monitoring_data <- xts(  
  cbind(  
    DetectedVulnerabilities = c(4, 6, 5, 8, 10, 7, 6, 5, 3, 2),  
    SecurityAlerts = c(3, 5, 4, 7, 9, 6, 5, 4, 3, 2),  
    ResolvedVulnerabilities = c(1, 2, 2, 3, 4, 4, 5, 5, 4, 3)  
  ),  
  order.by = days  
)  
  
dygraph(  
  monitoring_data,  
  main = "Continuous Security Monitoring and Vulnerability Detection",  
  xlab = "Monitoring Date",  
  ylab = "Security Events"
```

) %>%

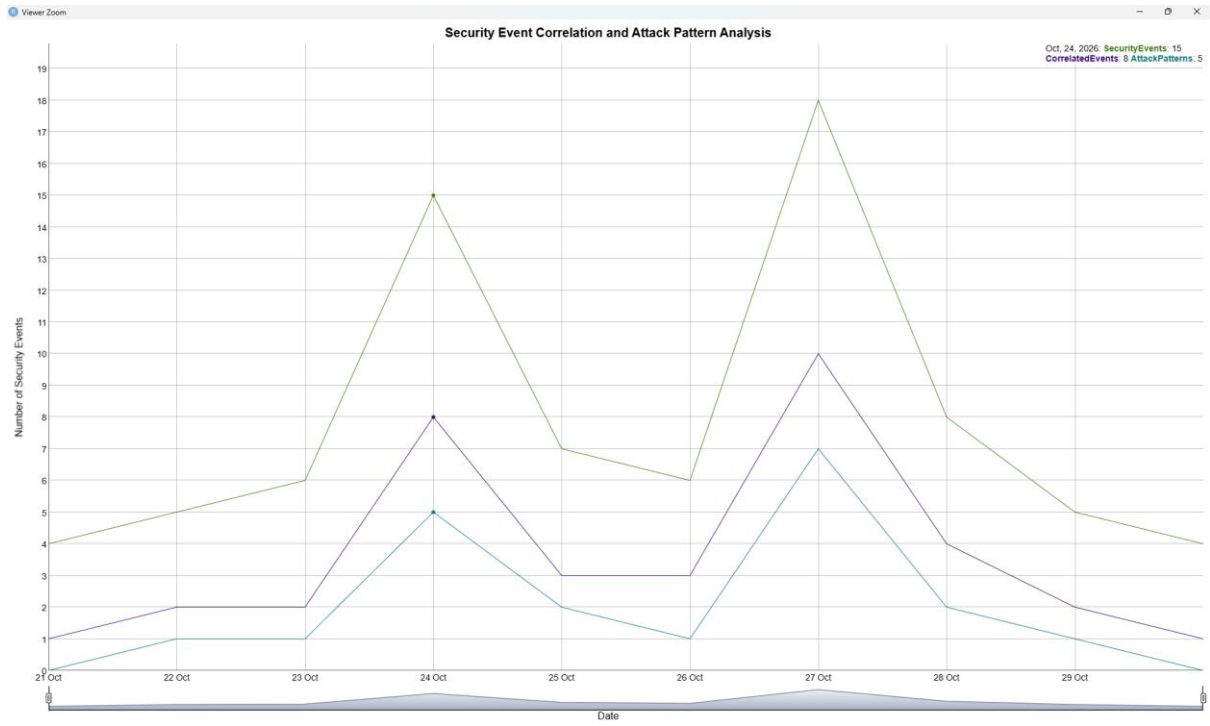
dyRangeSelector()



```
# =====  
# EXPERIMENT 29  
# Security Event Correlation and Attack Pattern Analysis  
# =====  
  
# Run once if required  
install.packages("dygraphs")  
install.packages("xts")  
  
library(dygraphs)  
library(xts)  
  
# Synthetic security event correlation data  
days <- as.Date("2026-10-21") + 0:9  
  
event_data <- xts(  
  cbind(  
    SecurityEvents = c(4, 5, 6, 15, 7, 6, 18, 8, 5, 4),  
    CorrelatedEvents = c(1, 2, 2, 8, 3, 3, 10, 4, 2, 1),  
    AttackPatterns = c(0, 1, 1, 5, 2, 1, 7, 2, 1, 0)  
  ),  
  order.by = days  
)  
  
dygraph(  
  event_data,  
  main = "Security Event Correlation and Attack Pattern Analysis",  
  xlab = "Date",  
  ylab = "Number of Security Events"
```

) %>%

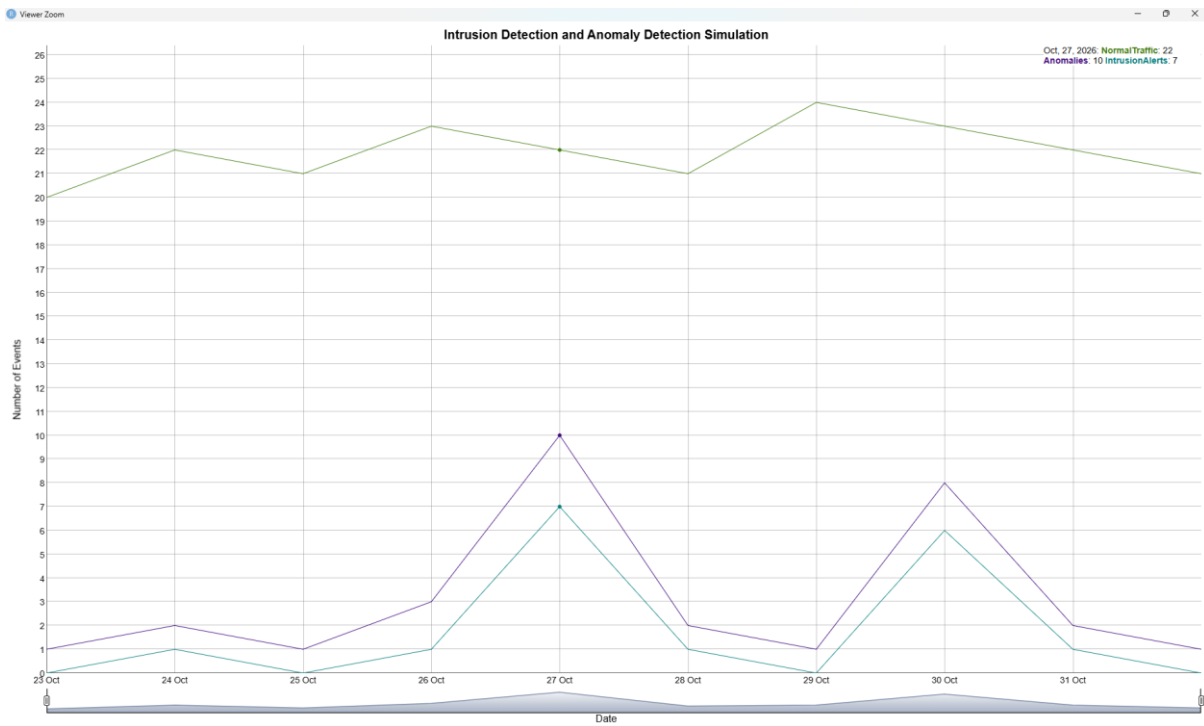
dyRangeSelector()



```
# =====  
# EXPERIMENT 30  
# Intrusion Detection and Anomaly Detection Simulation  
# =====  
  
# Run once if required  
install.packages("dygraphs")  
install.packages("xts")  
  
library(dygraphs)  
library(xts)  
  
# Synthetic intrusion detection data  
days <- as.Date("2026-10-23") + 0:9  
  
intrusion_data <- xts(  
  cbind(  
    NormalTraffic = c(20, 22, 21, 23, 22, 21, 24, 23, 22, 21),  
    Anomalies = c(1, 2, 1, 3, 10, 2, 1, 8, 2, 1),  
    IntrusionAlerts = c(0, 1, 0, 1, 7, 1, 0, 6, 1, 0)  
  ),  
  order.by = days  
)  
  
dygraph(  
  intrusion_data,  
  main = "Intrusion Detection and Anomaly Detection Simulation",  
  xlab = "Date",  
  ylab = "Number of Events"
```

) %>%

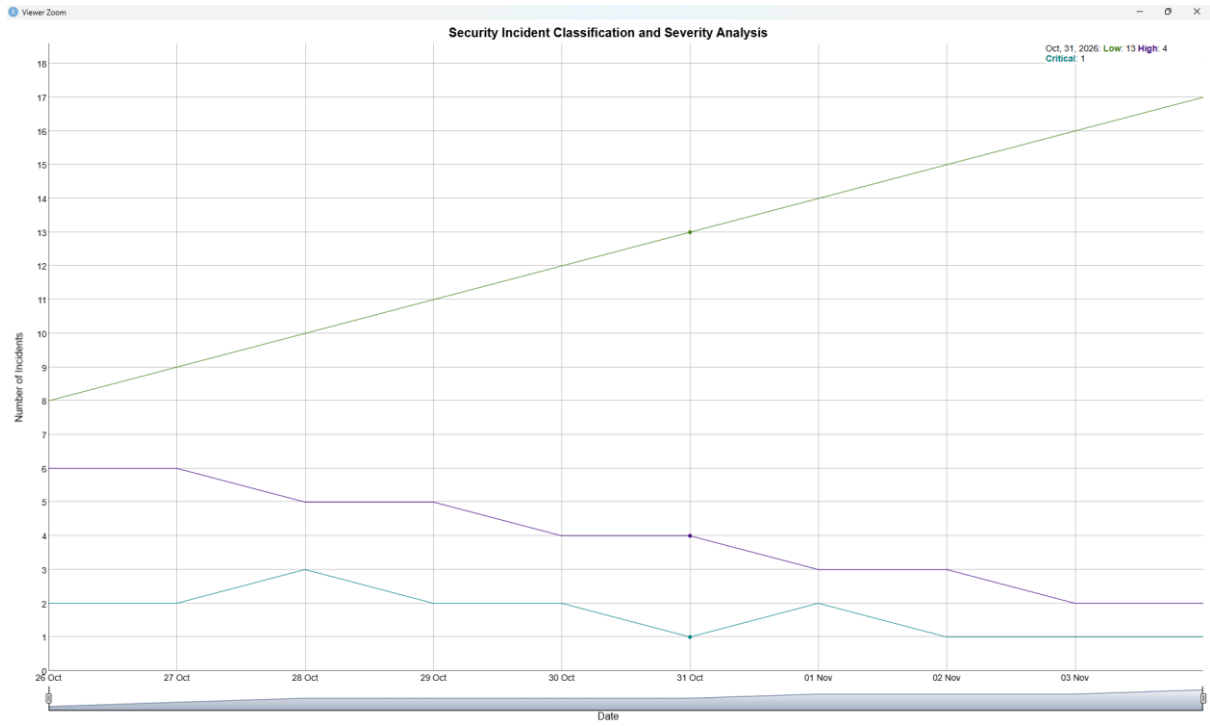
dyRangeSelector()



```
# =====  
# EXPERIMENT 31  
# Security Incident Classification and Severity Analysis  
# =====  
  
# Run once if required  
install.packages("dygraphs")  
install.packages("xts")  
  
library(dygraphs)  
library(xts)  
  
# Synthetic incident severity data  
days <- as.Date("2026-10-26") + 0:9  
  
incident_data <- xts(  
  cbind(  
    Low = c(8, 9, 10, 11, 12, 13, 14, 15, 16, 17),  
    High = c(6, 6, 5, 5, 4, 4, 3, 3, 2, 2),  
    Critical = c(2, 2, 3, 2, 2, 1, 2, 1, 1, 1)  
  ),  
  order.by = days  
)  
  
dygraph(  
  incident_data,  
  main = "Security Incident Classification and Severity Analysis",  
  xlab = "Date",  
  ylab = "Number of Incidents"
```


) %>%

dyRangeSelector()



```
# =====  
# EXPERIMENT 32  
# Incident Response Workflow Modelling and Simulation  
# =====
```

```
# Run once if required  
install.packages("DiagrammeR")
```

```
library(DiagrammeR)
```

```
grViz("  
digraph IncidentResponse {
```

```
graph [  
  rankdir=LR,  
  bgcolor=white,  
  label='Incident Response Workflow',  
  labelloc=t,  
  fontsize=20  
]
```

```
node [  
  shape=box,  
  style='rounded, filled',  
  fontname=Helvetica,  
  fontsize=13  
]
```

```
Detect [  
  shape=oval,  
  style='filled',  
  fontname=Helvetica,  
  fontsize=13  
]
```

```
    label='Detect\\nSecurity Incident',  
    fillcolor=lightblue  
]
```

```
Classify [  
    label='Classify\\nIncident',  
    fillcolor=lightyellow  
]
```

```
Contain [  
    label='Contain\\nThreat',  
    fillcolor=lightpink  
]
```

```
Eradicate [  
    label='Eradicate\\nRoot Cause',  
    fillcolor=lightgreen  
]
```

```
Recover [  
    label='Recover\\nSystem',  
    fillcolor=lightcyan  
]
```

```
Monitor [  
    label='Monitor\\n& Verify',  
    fillcolor=lavender  
]
```

```
Report [
    label='Incident Report\\n& Lessons Learned',
    fillcolor=wheat
]
```

Detect -> Classify

Classify -> Contain

Contain -> Eradicate

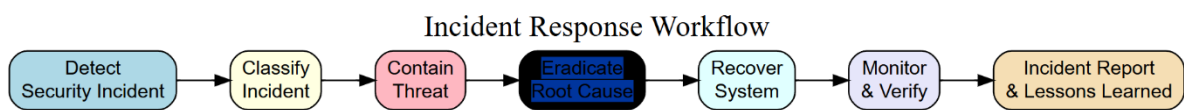
Eradicate -> Recover

Recover -> Monitor

Monitor -> Report

```
}
")
```

Viewer Zoom



=====

EXPERIMENT 33

```
# Security Incident Timeline Reconstruction and Root Cause Analysis
```

```
# =====
```

```
# Run once if required
```

```
install.packages("dygraphs")
```

```
install.packages("xts")
```

```
library(dygraphs)
```

```
library(xts)
```

```
# Synthetic incident timeline data
```

```
days <- as.Date("2026-10-29") + 0:9
```

```
incident_data <- xts(
```

```
  cbind(
```

```
    IncidentEvents = c(1, 2, 4, 8, 12, 9, 5, 3, 2, 1),
```

```
    SuspiciousEvents = c(0, 1, 2, 5, 9, 6, 3, 2, 1, 0),
```

```
    ResolvedEvents = c(0, 0, 1, 2, 3, 5, 6, 7, 8, 9)
```

```
  ),
```

```
  order.by = days
```

```
)
```

```
dygraph(
```

```
  incident_data,
```

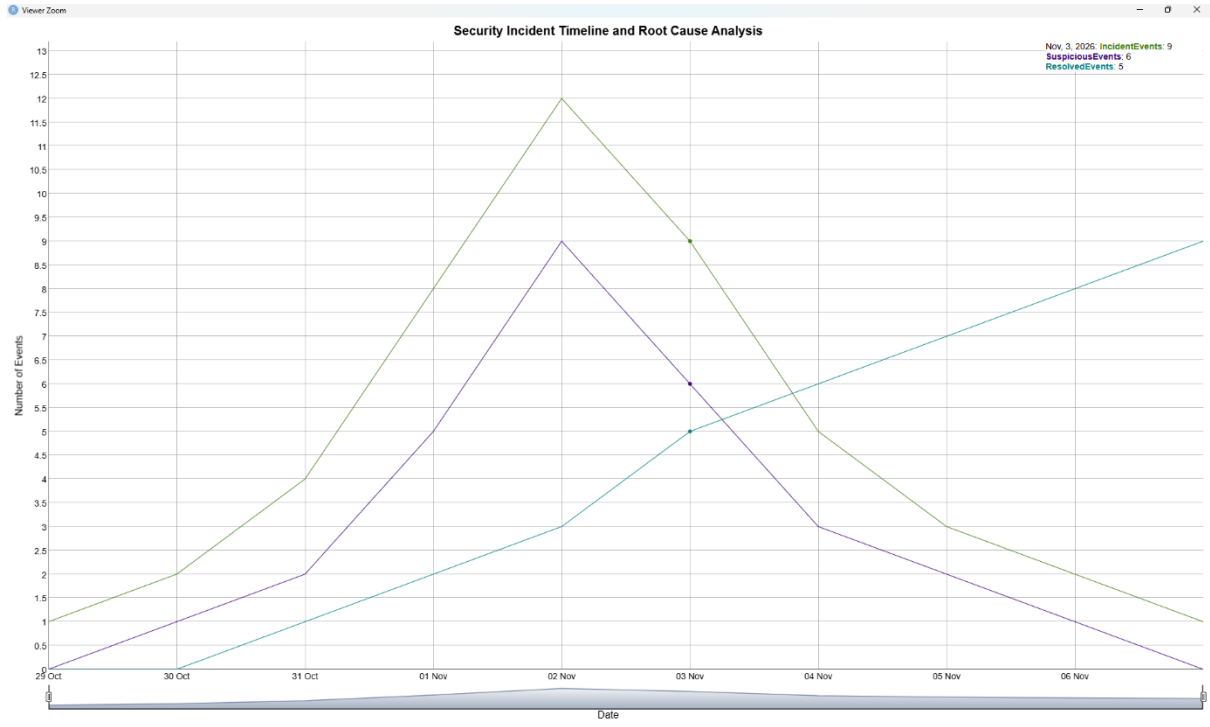
```
  main = "Security Incident Timeline and Root Cause Analysis",
```

```
  xlab = "Date",
```

```
  ylab = "Number of Events"
```

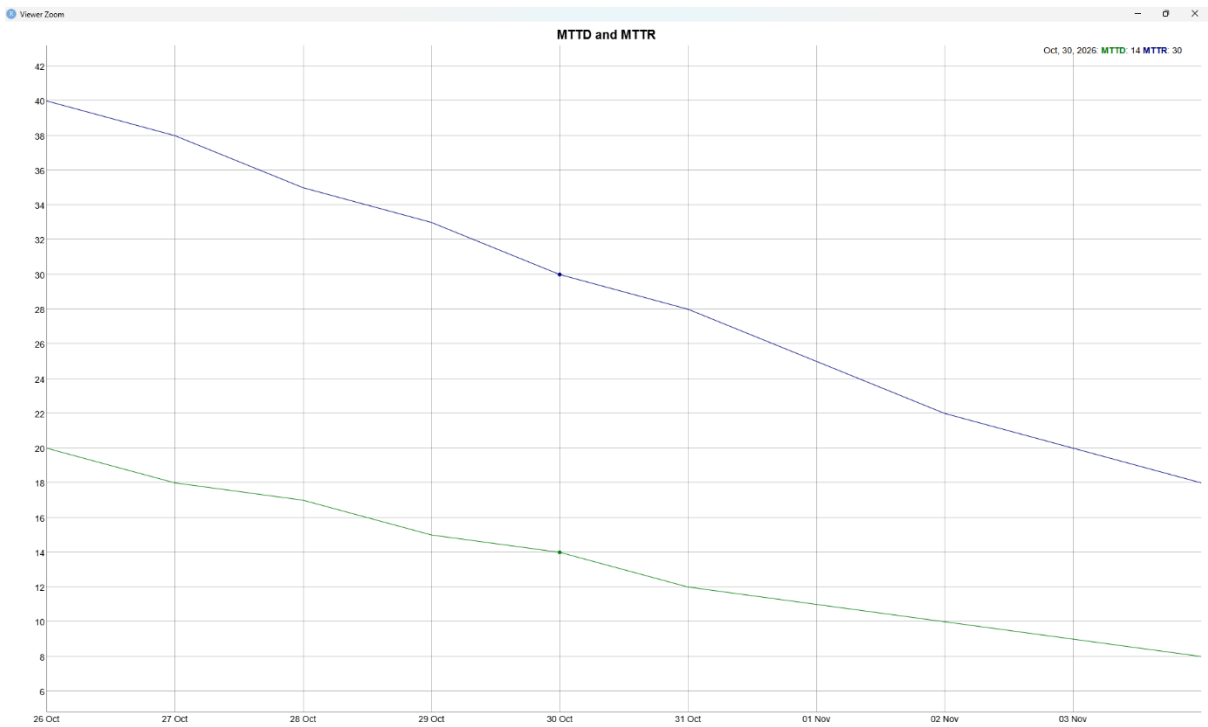
```
) %>%
```

```
  dyRangeSelector()
```



```
# =====  
# EXPERIMENT 34  
# MTTD and MTTR Analysis  
# =====  
  
# Run once if required  
install.packages("dygraphs")  
install.packages("xts")  
  
library(dygraphs)  
library(xts)  
  
# Synthetic MTTD and MTTR data  
days <- as.Date("2026-11-02") + 0:9  
  
incident_data <- xts(  
  cbind(  
    MTTD = c(20, 18, 17, 15, 14, 12, 11, 10, 9, 8),  
    MTTR = c(40, 38, 35, 33, 30, 28, 25, 22, 20, 18)  
  ),  
  order.by = days  
)  
  
dygraph(  
  incident_data,  
  main = "MTTD and MTTR Analysis",  
  xlab = "Date",  
  ylab = "Time (Minutes)"  
) %>%
```

dyRangeSelector()




```
# =====
# EXPERIMENT 35
# Security Compliance Monitoring and Governance Simulation
# =====

# Run once if required
install.packages("dygraphs")
install.packages("xts")

library(dygraphs)
library(xts)

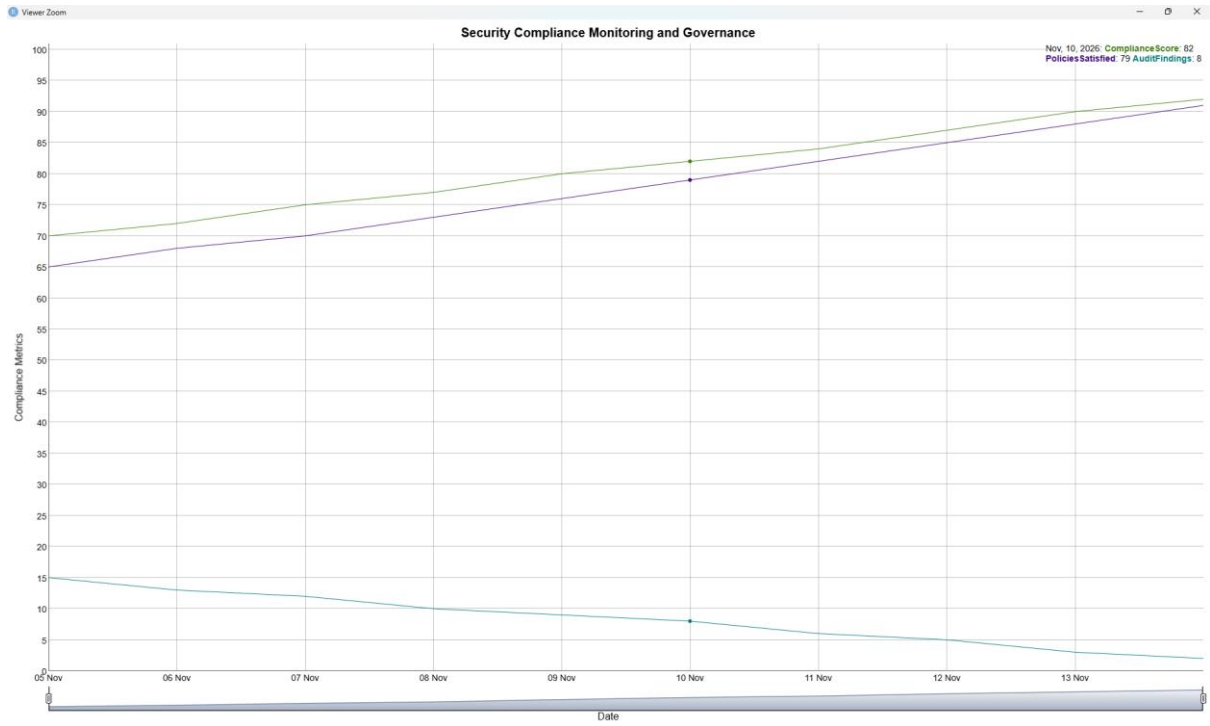
# Synthetic security compliance data
days <- as.Date("2026-11-05") + 0:9

compliance_data <- xts(
  cbind(
    ComplianceScore = c(70, 72, 75, 77, 80, 82, 84, 87, 90, 92),
    PoliciesSatisfied = c(65, 68, 70, 73, 76, 79, 82, 85, 88, 91),
    AuditFindings = c(15, 13, 12, 10, 9, 8, 6, 5, 3, 2)
  ),
  order.by = days
)

dygraph(
  compliance_data,
  main = "Security Compliance Monitoring and Governance",
  xlab = "Date",
  ylab = "Compliance Metrics"
```

) %>%

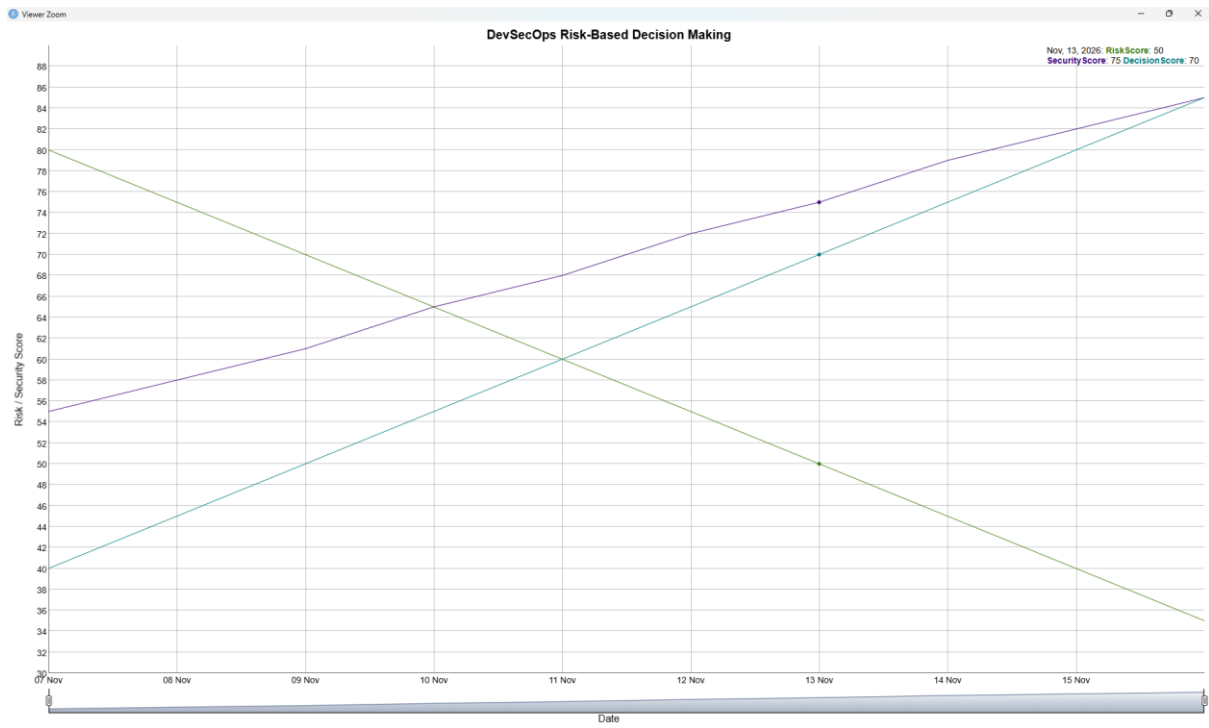
dyRangeSelector()



```
# =====  
# EXPERIMENT 36  
# DevSecOps Risk-Based Decision Making using Synthetic Data  
# =====  
  
# Run once if required  
install.packages("dygraphs")  
install.packages("xts")  
  
library(dygraphs)  
library(xts)  
  
# Synthetic risk-based decision data  
days <- as.Date("2026-11-07") + 0:9  
  
risk_data <- xts(  
  cbind(  
    RiskScore = c(80, 75, 70, 65, 60, 55, 50, 45, 40, 35),  
    SecurityScore = c(55, 58, 61, 65, 68, 72, 75, 79, 82, 85),  
    DecisionScore = c(40, 45, 50, 55, 60, 65, 70, 75, 80, 85)  
  ),  
  order.by = days  
)  
  
dygraph(  
  risk_data,  
  main = "DevSecOps Risk-Based Decision Making",  
  xlab = "Date",  
  ylab = "Risk / Security Score"
```

) %>%

dyRangeSelector()



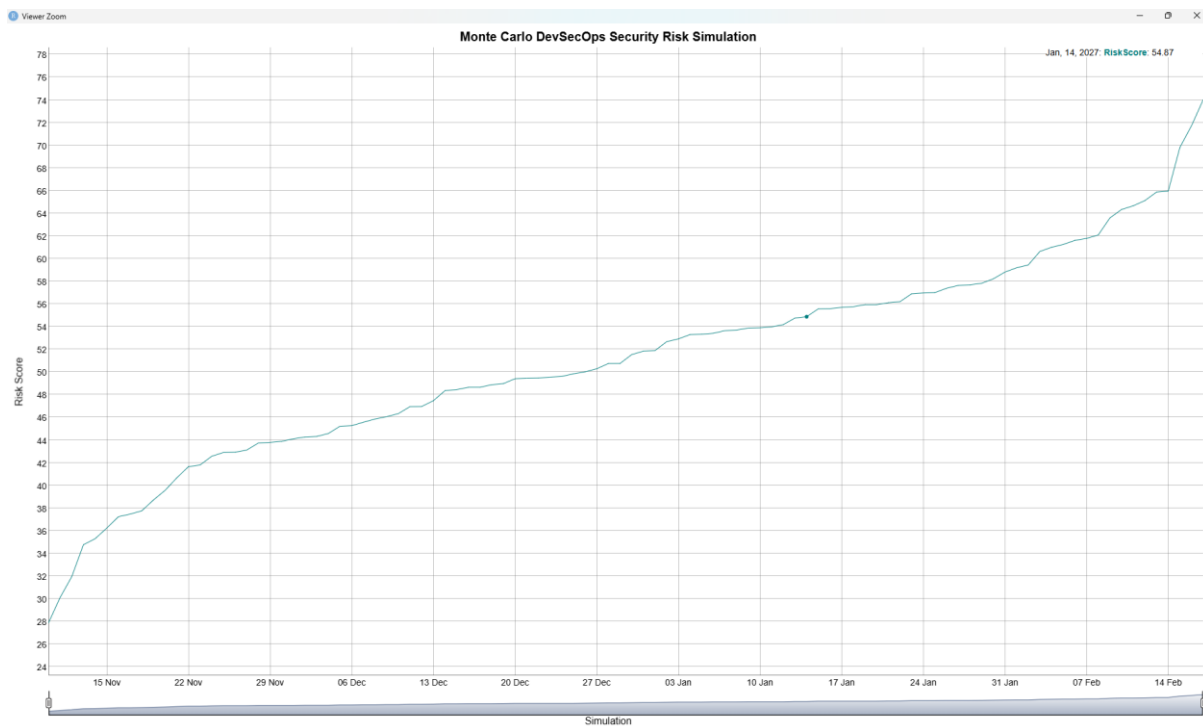
```
# =====  
# EXPERIMENT 37  
# Monte Carlo Simulation for DevSecOps Security Risk Analysis  
# =====  
  
# Run once if required  
install.packages("dygraphs")  
install.packages("xts")  
  
library(dygraphs)  
library(xts)  
  
# Generate Monte Carlo risk simulation data  
set.seed(1)  
  
simulations <- 100  
risk <- rnorm(simulations, mean = 50, sd = 10)  
  
# Keep risk values within a practical range  
risk <- pmax(0, pmin(100, risk))  
  
# Sort risk values for clear visualization  
risk <- sort(risk)  
  
days <- as.Date("2026-11-10") + 0:(simulations - 1)  
  
risk_data <- xts(  
  cbind(  
    RiskScore = risk
```

```

),
order.by = days
)

dygraph(
  risk_data,
  main = "Monte Carlo DevSecOps Security Risk Simulation",
  xlab = "Simulation",
  ylab = "Risk Score"
) %>%
  dyRangeSelector()

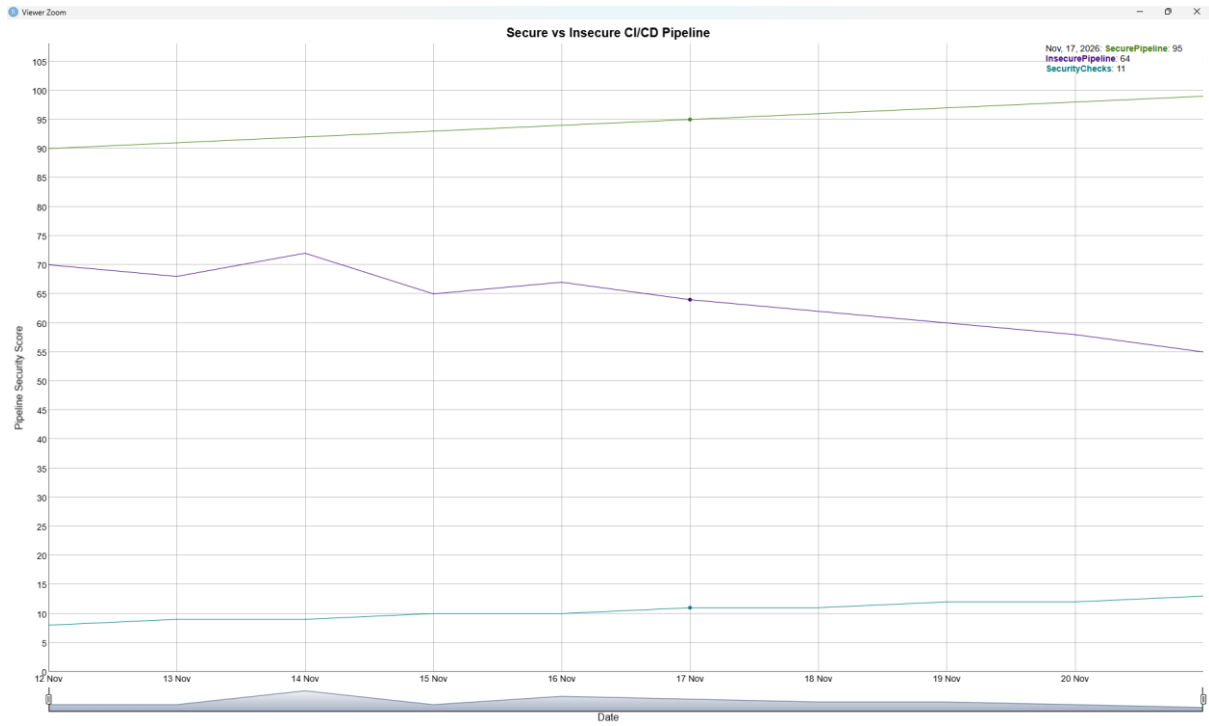
```



```
# =====  
# EXPERIMENT 38  
# Comparative Evaluation of Secure and Insecure CI/CD Pipelines  
# =====  
  
# Run once if required  
install.packages("dygraphs")  
install.packages("xts")  
  
library(dygraphs)  
library(xts)  
  
# Synthetic CI/CD pipeline comparison data  
days <- as.Date("2026-11-12") + 0:9  
  
pipeline_data <- xts(  
  cbind(  
    SecurePipeline = c(90, 91, 92, 93, 94, 95, 96, 97, 98, 99),  
    InsecurePipeline = c(70, 68, 72, 65, 67, 64, 62, 60, 58, 55),  
    SecurityChecks = c(8, 9, 9, 10, 10, 11, 11, 12, 12, 13)  
  ),  
  order.by = days  
)  
  
dygraph(  
  pipeline_data,  
  main = "Secure vs Insecure CI/CD Pipeline",  
  xlab = "Date",  
  ylab = "Pipeline Security Score"
```

) %>%

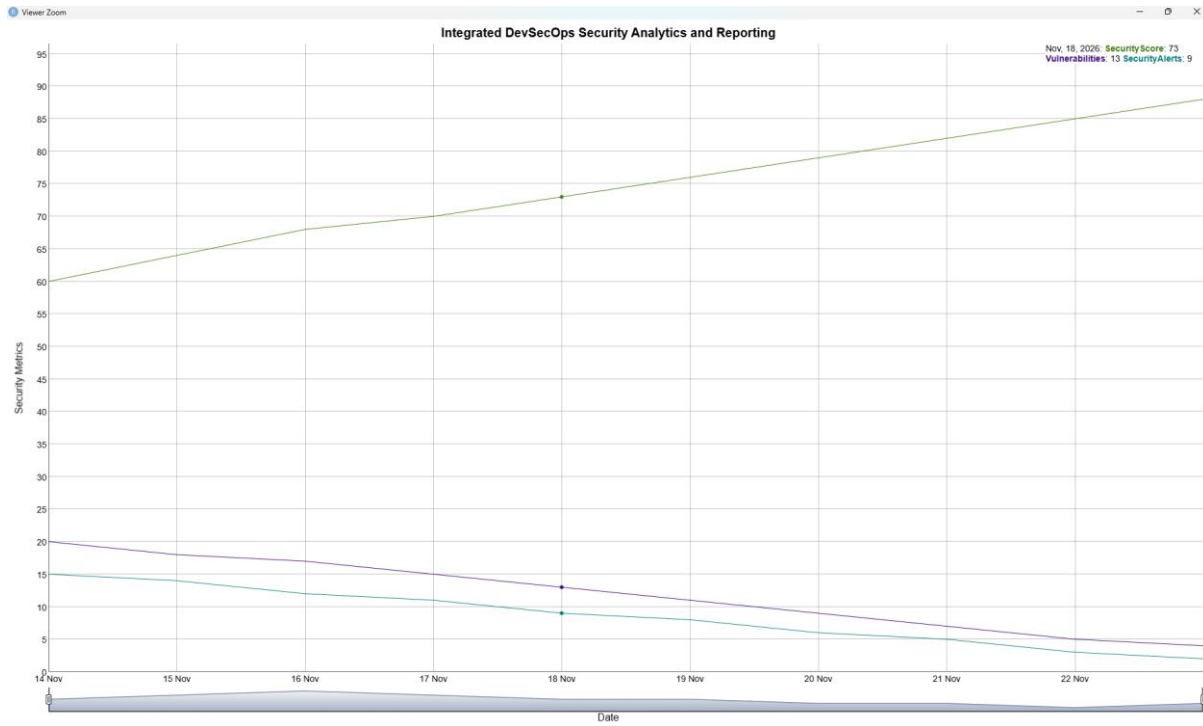
dyRangeSelector()




```
# =====  
# EXPERIMENT 39  
# Integrated DevSecOps Security Analytics and Reporting  
# =====  
  
# Run once if required  
install.packages("dygraphs")  
install.packages("xts")  
  
library(dygraphs)  
library(xts)  
  
# Synthetic integrated security analytics data  
days <- as.Date("2026-11-14") + 0:9  
  
security_data <- xts(  
  cbind(  
    SecurityScore = c(60, 64, 68, 70, 73, 76, 79, 82, 85, 88),  
    Vulnerabilities = c(20, 18, 17, 15, 13, 11, 9, 7, 5, 4),  
    SecurityAlerts = c(15, 14, 12, 11, 9, 8, 6, 5, 3, 2)  
  ),  
  order.by = days  
)  
  
dygraph(  
  security_data,  
  main = "Integrated DevSecOps Security Analytics and Reporting",  
  xlab = "Date",  
  ylab = "Security Metrics"
```

) %>%

dyRangeSelector()



```
# =====  
# EXPERIMENT 40  
# End-to-End DevSecOps Pipeline Security Simulation and Evaluation  
# =====  
  
# Run once if required  
install.packages("dygraphs")  
install.packages("xts")  
  
library(dygraphs)  
library(xts)  
  
# Synthetic end-to-end DevSecOps pipeline data  
days <- as.Date("2026-11-16") + 0:9  
  
pipeline_data <- xts(  
  cbind(  
    Build = c(70, 72, 75, 78, 80, 82, 84, 86, 88, 90),  
    Security = c(60, 64, 68, 72, 75, 78, 81, 84, 87, 90),  
    Deployment = c(55, 60, 65, 68, 72, 76, 80, 84, 87, 91)  
  ),  
  order.by = days  
)  
  
dygraph(  
  pipeline_data,  
  main = "End-to-End DevSecOps Pipeline Security Simulation",  
  xlab = "Date",  
  ylab = "Pipeline Score"
```

) %>%

dyRangeSelector()

